

Chapter 146 — Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

- **Small, deceptively simple glands** (formerly the suprarenal glands) at an anatomical crossroads beside major vessels, nerves and viscera; now recognised as **central to homeostasis** and responsible for several major human diseases.
- **Medical management is multidisciplinary** (endocrinology, nephrology, cardiology); surgical ownership has historically been split between urologists, general, oncological and endocrine surgeons, largely by **local institutional tradition**.
- Evaluation and surgery **sit naturally with urologists** because of minimally invasive skill, familiarity with retroperitoneal anatomy and approaches, and the **close linkage of renal and adrenal pathophysiology**.

HISTORICAL BACKGROUND

- **Overlooked by Galen, da Vinci and Vesalius**; first described by **Bartholomaeus Eustachius** (mid-16th century).
- **Thomas Addison (1855)** linked adrenal insufficiency to the glands at autopsy — the eponymous Addison disease.
- **Brown-Séquard (1856)**: bilateral adrenalectomy in animals was uniformly fatal — the adrenals are **indispensable to life**.
- **Osler (1896)**: first hormonal replacement — crude porcine adrenal extract produced weight gain in one patient with Addison disease.
- **Adrenaline** discovered and localised to the medulla; shown to produce a sustained rise in blood pressure. Its **failure to sustain life after bilateral adrenalectomy** established Addison disease as a disorder of the **cortex alone**.
- **Cortisol** isolated in the 1930s; its use in rheumatoid arthritis earned the **1950 Nobel Prize** (Kendall, Hench, Reichstein). **Aldosterone** isolated from bovine adrenal in **1952**.
- Late 20th-century pioneers: **Conn, Wilkins, Liddle, Sutherland**.

ADRENAL EMBRYOLOGY AND ANATOMY

Overview

- **Paired retroperitoneal organs of cortex and medulla**: spiculated **mustard-yellow cortex**, distinct from perinephric fat; **brown central medulla** on cross-section.
- **4–5 g each; 4–6 cm long, 2–3 cm wide. Right triangular, left crescent-shaped.**
- Position either “**capping**” the upper pole or “**cradled**” **superomedially** just above the renal vessels — visible on computed tomography (CT) or magnetic resonance imaging (MRI) and **surgically relevant**, especially on the left and in patients with abundant perinephric fat, when identifying the adrenal vein.

Embryology

CORTX VERSUS MEDULLA: EMBRYOLOGICAL ORIGIN

	Cortex	Medulla
Origin	Intermediate mesoderm of the urogenital ridge	Neural crest cells from adjacent sympathetic ganglia

	Cortex	Medulla
Timeline	Week 5 fetal cortex forms → weeks 6–7 mesothelial cells surround it (future adult cortex) → end of week 8 encapsulated and separated from peritoneal mesothelium → adult cells replace fetal cells	Migrate into the medial fetal cortex by week 9 → central position around the adrenal vein by week 18
Function	Endocrine (steroids)	Neurocrine (catecholamines)

- The medullary migration through the cortex explains the **brown stippling of the cortex**.
- **Fetal adrenal at birth is twice adult weight** but immature; the fetal cortex is **resorbed by 12 months**. Glomerulosa and fasciculata continue to develop, but the **zona reticularis completes differentiation only at 3 years** — reflecting the late importance of adrenal sex steroids. Adrenocortical stem cells may contribute to development and homeostasis.
- **Unilateral adrenal agenesis is rare**; its reported association with renal agenesis is **probably spurious** (poor imaging of the ipsilateral adrenal) because adrenal and renal development are separate. With renal agenesis, malrotation or malascence the adrenal usually develops normally, **discoid and in its normal position**.
- **Adrenal rests** (accessory tissue, cortical or medullary) lie along the **path of gonadal descent**: present in up to half of neonates, persisting in about 1% of adults.
- In **congenital adrenal hyperplasia (CAH)**, **testicular adrenal rests may become hyperplastic and present as a testicular mass** — consider before orchiectomy in CAH.
- **Adrenal heterotopia**: incomplete separation of adrenal mesoderm from liver or kidney, with partial or complete incorporation into that organ.

Anatomy

RIGHT VERSUS LEFT ADRENAL: RELATIONS AND VESSELS

	Right	Left
Shared	Within Gerota's fascia at 11th–12th ribs; cephalad to upper pole; anterior to diaphragmatic crus	
Medial	Inferior vena cava (IVC)	Aorta
Anterior	Liver	Stomach, pancreas, splenic vessels
Adrenal vein	Short, directly into IVC	Long; joined by inferior phrenic vein → left renal vein
Lymphatics	Paracaval nodes	Para-aortic nodes

- **Adjacent-organ lesions can mimic an adrenal mass** (e.g., gastric leiomyoma or greater-curve diverticulum).
- **Redundant arterial supply** from three sources: **superior** (inferior phrenic artery), **middle** (aorta directly), **inferior** (ipsilateral renal artery) → subcapsular plexus → direct medullary branches and cortical sinusoids.
- **Venous drainage is more predictable than arterial**: medullary veins coalesce into a central adrenal vein surrounded by medulla.
- **Overlapping arterial and venous anatomy permits partial adrenalectomy** with little risk of infarction.
- **Anomalous venous drainage in up to 9%** — important for adrenal vein sampling (AVS).
- **Innervation**: **preganglionic sympathetic fibres go directly to medullary chromaffin cells**; postganglionic fibres from splanchnic ganglia supply the cortex; parasympathetic (vagal) supply is ill defined.

Figure legend — Fig. 146.3

- A “left adrenal mass” abutting the stomach proved on CT, MRI and metaiodobenzylguanidine (MIBG) imaging to be separate from a normal adrenal: **a gastric neuroendocrine tumour**, treated by partial gastrectomy — a high index of suspicion prevents operative misadventure.

Histology

ZONES OF THE ADRENAL CORTEX

Zone (outer → inner)	Cells	Main product	Daily output
Glomerulosa	Small polyhedral, scant eosinophilic cytoplasm, dark round nuclei	Mineralocorticoid (aldosterone)	100–150 µg
Fasciculata	Broad layer of large pale cells in vertical columns	Glucocorticoid (cortisol)	10–20 mg
Reticularis	Round dark-staining cells	Sex steroids (androgens, oestrogens)	>20 mg

- Each gland lies within a **fibrous capsule**; the cortex lies directly beneath it.

KEY POINTS: ANATOMY AND EMBRYOLOGY

- **Two embryologically and functionally distinct components:** cortex (endocrine) and medulla (neurocrine).
- **Arterial supply variable; venous drainage predictable** — short right vein to IVC, longer left vein to left renal vein in most patients.

Concept — The adrenal is two organs sharing a capsule: a mesodermal steroid factory and a neural-crest sympathetic ganglion. Its surgical anatomy is governed by one asymmetry — the short right vein on the IVC versus the long left vein on the renal vein.

ADRENAL PHYSIOLOGY

- **Cortex and medulla are best conceived as two distinct organs** — the former endocrine, the latter neurocrine.

Adrenal Cortex Physiology

- **All cortical steroids derive from cholesterol**, chiefly from **low-density lipoprotein (LDL)**; the **ratio and type of enzymes in each zone** determine its product (Fig. 146.4 — steroid synthesis from cholesterol to mineralocorticoid, glucocorticoid and androgen).
- **No membrane receptors:** lipophilic steroids diffuse into the cell, bind **intracellular receptors**, and the hormone–receptor complex binds DNA to modulate transcription.

Zona Glomerulosa

- **The only zone expressing aldosterone synthase (CYP11B2)** — hence the sole source of aldosterone.
- Aldosterone acts on the **distal nephron**: reabsorbs Na^+ and Cl^- , secretes K^+ and H^+ . Sodium reabsorption carries free water, so aldosterone changes **total body fluid volume, not sodium concentration**. Salivary and colonic effects are physiologically minor.
- **Regulated chiefly by angiotensin II (renin–angiotensin–aldosterone system, RAAS) and directly by serum potassium**; adrenocorticotrophic hormone (ACTH) is a weak stimulus — so the **glomerulosa alone does not atrophy in pituitary failure**.
- **Atrial natriuretic peptide is the main inhibitor** (with somatostatin, dopamine) — a cardiac–adrenal–renal link.

TABLE 146.1 PRIMARY EFFECTS OF MINERALOCORTICIDS

Action	Effect	Site
Renal Na ⁺ reabsorption	↑ blood volume, ↑ blood pressure, ↓ urine sodium	Distal tubule, connecting segment, collecting duct
Renal Cl ⁻ reabsorption	↑ serum chloride, ↓ urine chloride	
Renal K ⁺ secretion	↓ serum potassium, ↑ urine potassium	
Renal proton secretion	↑ urine ammonium	Collecting duct (consequence of Na ⁺ reabsorption)

- **What it tests:** every mineralocorticoid effect is a distal-nephron exchange driven by sodium reabsorption.
Trap: expecting hyponatraemia — the volume, not the concentration, rises.

Zona Fasciculata

- **Glucocorticoid synthesis via 17 α -, 21- and 11 β -hydroxylase.** Cortisol is the principal human glucocorticoid, under **tight ACTH control** in a classic negative-feedback loop.
- **Strict circadian rhythm:** most cortisol secreted in the early morning. Glucocorticoids are **essential to life** — metabolism, immunity, intravascular volume, blood pressure, mood, sleep and possibly memory.

Normal hypothalamic–pituitary–adrenal (HPA) axis physiology

- Fasciculata secretes **up to 20 mg cortisol per day**.
- Hypothalamic **corticotropin-releasing hormone (CRH)** (with oxytocin and vasopressin) → anterior pituitary corticotrophs cleave **pro-opiomelanocortin (POMC)** to ACTH → adrenal cortisol and androgens → cortisol feeds back on hypothalamus (CRH) and pituitary (ACTH). Fig. 146.5 depicts the loop.
- **Stress, physiological or psychological, is the dominant modulator;** splanchnic innervation also influences output.
- **ACTH is trophic as well as secretagogue:** without it (e.g., suppression by exogenous steroid) all but the mineralocorticoid-producing cells stop synthesis and **undergo apoptosis**.
- CRH follows the **suprachiasmatic nucleus: cortisol peaks in the morning, nadir about 11 pm;** even small perturbations of this rhythm are pathological.

TABLE 146.2 PRIMARY EFFECTS OF GLUCOCORTICOIDS

Effect	Clinical implication
Enhance skeletal and cardiac contraction	Deficiency → weakness
Protein catabolism	Excess → muscle wasting, weakness
Inhibit bone formation	Excess → reduced bone density
Inhibit collagen synthesis	Excess → thin skin, fragile capillaries
↑ vascular contractility, ↓ permeability	Deficiency → difficulty maintaining blood pressure
Anti-inflammatory; anti-immune	Exogenous steroid treats inflammatory and autoimmune disease; promotes transplant tolerance
Maintain glomerular filtration rate (GFR)	Deficiency → reduced GFR

From Howards SS, Carey RM, 1991.

- **Pattern:** excess produces the catabolic cushingoid picture (wasting, osteoporosis, thin skin); deficiency produces the addisonian picture (weakness, hypotension, low GFR).

Zona Reticularis

- **17 α -hydroxylase and 17,20-lyase** → **dehydroepiandrosterone (DHEA), sulphated DHEA (DHEA-S) and androstenedione**; some androgen also from the fasciculata. ACTH-controlled and circadian.
- **The largest steroid output (>20 mg/day) but the least important for adult homeostasis.**
- Pharmacological suppression of adrenal androgens is an **increasingly used strategy in advanced prostate cancer**; developmental aberrations cause **CAH**.

Adrenal Medulla Physiology

- **<10% of adrenal mass**; unrelated to the cortex in function and embryology; an **integral part of the autonomic nervous system**.
- **Chromaffin cells receive preganglionic sympathetic fibres from T11–L2** — analogous to sympathetic ganglion cells.
- Secretes **epinephrine 80%, norepinephrine 19%, dopamine 1%** — catecholamines synthesised from **tyrosine**, modulating the stress response via organ-specific adrenoceptors (Table 146.3). Stored in vesicles, released by **exocytosis**.
- **Phenylethanolamine-N-methyltransferase (PNMT)** converts norepinephrine to epinephrine and is **relatively unique to the medulla** (also brain and organ of Zuckerkandl). **Glucocorticoids potentiate PNMT** — one of the few cortex–medulla links — and explain why the medulla is the **main source of circulating epinephrine**.

Catecholamine metabolism (Fig. 146.6)

- Tyrosine → L-dihydroxyphenylalanine (L-DOPA) → dopamine → norepinephrine → (PNMT) → epinephrine.
- **Catechol-O-methyltransferase (COMT)** methylates norepinephrine → **normetanephrine** and epinephrine → **metanephrine**. Most adrenal metabolism occurs **within medullary cells themselves**, not in liver or kidney.
- **>90% of circulating metanephrine and $\geq 20\%$ of normetanephrine derive from the adrenal medulla** — so their rise is the basis of pheochromocytoma diagnosis. Urinary metabolites are largely **sulphated**.
- **Monoamine oxidase (MAO)** and other enzymes produce **vanillylmandelic acid (VMA)**, the end product, formed largely in the liver; sympathetic-nerve catecholamines also end as VMA.
- Terminology: “**free**” = unconjugated; “**fractionated**” = individual compounds reported separately.

TABLE 146.3 PRIMARY EFFECTS OF CATECHOLAMINES (GROUPED BY SYSTEM)

System	Receptors	Dominant response	Clinical manifestation
Eye	α_1 (radial iris); β_2 (ciliary)	Mydriasis; relaxation for far vision	Blurred vision
Heart (SA and AV node, atria, His–Purkinje, ventricles)	β_1 , β_2	$\uparrow$ rate, contractility, conduction, automaticity (strongest at AV node, conducting system, ventricles)	Palpitations, angina
Arterioles	α_1 , α_2 (constrict); β_2 (dilate)	Skin, mucosa, abdominal viscera, renal: strong constriction; coronary, pulmonary, muscle: mixed	Pallor, hypertension, angina, bowel ischaemia, renal failure, stroke, oedema
Systemic veins	α_1 , α_2 , β_2	Constriction and dilatation	Orthostatic hypotension
Lung	β_2 ; α_1	Bronchial relaxation; secretion altered	—
Gut, gallbladder	α , β_2 (tone $\downarrow$); α_1 (sphincters contract); α_2 (secretion $\downarrow$)	$\downarrow$ motility, sphincter contraction	Early satiety, constipation, ileus, gallstones

System	Receptors	Dominant response	Clinical manifestation
Kidney	$\alpha 1$ (↓), $\beta 2$ (↑)	Renin secretion	—
Bladder	$\beta 2$ (detrusor relaxes); $\alpha 1$ (trigone, sphincter contract)	Storage	Urinary retention
Ureter, uterus, male sex organs	$\alpha 1$ ($\pm$ $\beta 2$ uterus)	↑ ureteric tone; uterine contraction (pregnant) or relaxation; ejaculation	—
Skin	$\alpha 1$	Piloerection; localised sweating	Sweating
Metabolic (muscle, pancreas, fat)	$\beta 2$ (muscle glycogenolysis, K^+ uptake); $\alpha 2$ (↓ insulin); $\beta 2$ (↑ insulin); $\alpha 2$, $\beta 1$, $\beta 2$ (lipolysis)	Glycogenolysis, insulin suppression , lipolysis, thermogenesis	Hyperglycaemia, glycosuria, feeling warm
Glands and others	$\alpha 1$, β (salivary); α (lacrimal); β (pineal); $\beta 1$ (posterior pituitary)	Salivary K^+ , water and amylase; lacrimation; melatonin; antidiuretic hormone ↑	Lacrimation, reduced diuresis
Spleen capsule	$\alpha 1$, $\beta 2$	Contraction	—

Modified from Pacak K, 2007. Source grades intensity from +++ (highest) to + (lowest).

- **What it tests:** the pheochromocytoma symptom complex is predictable from receptor maps — $\alpha 1$ vasoconstriction (hypertension, pallor, visceral ischaemia), $\beta 1$ cardiac stimulation (palpitations, angina), $\alpha 2$ insulin suppression (hyperglycaemia). **Trap:** the $\alpha 2$ -mediated insulin suppression is what rebounds as **post-resection hypoglycaemia**.

KEY POINTS: PHYSIOLOGY

- **Glomerulosa** → mineralocorticoid, fasciculata → glucocorticoid, reticularis → androgen; zonal enzyme ratios determine the product.
- **Aldosterone is driven by angiotensin II and potassium;** ACTH is minor, so the glomerulosa survives pituitary failure.
- **Cortisol is under tight ACTH control;** adrenal androgen suppression is a target in advanced prostate cancer.
- **Catecholamines derive from tyrosine** and modulate the systemic stress response.

Concept — Each zone is defined by which enzymes it expresses and which signal it obeys: the glomerulosa answers to angiotensin II and potassium, the inner zones to ACTH. That single split explains why pituitary failure spares aldosterone, and why intra-adrenal methylation makes metanephrines the diagnostic currency of the medulla.

EVALUATION OF ADRENAL LESIONS IN UROLOGICAL PRACTICE

- Urologists meet adrenal pathology as a **referred new mass**, an **incidental mass in an existing patient**, or occasionally a **clinically suspected syndrome** before imaging.
- **Two questions drive management:** (1) could the lesion be **malignant**? (2) is it **hormonally active** — secreting outside normal regulatory control? Either → treatment, usually resection (Box 146.1).
- **Malignant potential is judged by imaging; hormonal activity by laboratory testing; biopsy helps only in rare scenarios.**
- The **pregnant patient with an adrenal mass** needs a tailored multidisciplinary plan (outside the chapter's scope).

Overview

BOX 146.1 SUMMARY OF SURGICAL INDICATIONS FOR ADRENALECTOMY

Category	Indication
Functional mass	Autonomous cortisol secretion; pheochromocytoma; aldosterone hypersecretion
Oncological risk	Mass >4 cm (except myelolipoma); malignant imaging features (lipid-poor, heterogeneous, irregular borders, infiltrating); incidentaloma growing >1 cm on follow-up; isolated adrenal metastasis (multidisciplinary decision)
Myelolipoma	Extremely large and/or symptomatic
During nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal, or not visualised because of large tumour; venous thrombus to the level of the adrenal vein
Bilateral adrenalectomy	Failed neurosurgical treatment of Cushing disease; selected ectopic ACTH syndrome; ACTH-independent macronodular adrenal hyperplasia (AIMAH); primary pigmented nodular adrenocortical disease (PPNAD)

- **Pattern:** every indication is either **function** or **cancer risk**. **Trap:** myelolipoma is the named exception to the 4-cm rule.

Imaging of Adrenal Masses

- **Imaging is the primary tool for malignant potential;** several guidelines address it.
- **No imaging feature reliably separates functional from non-functional adenomas** — endocrine testing can never be omitted on imaging grounds.

Intracellular Fat Content

- **Benign adenomas are lipid-rich** and can often be diagnosed on this feature alone.

Unenhanced computed tomography

- **The first and perhaps single best, most easily interpreted test** for intracellular lipid; diagnoses adenoma in >70% of cases.
- **<10 Hounsfield units (HU) = high intracytoplasmic lipid = adenoma** in patients without another known malignancy.
- **≤0 HU is 100% specific but insensitive** (a minority of adenomas reach it) — hence the working **10-HU threshold**: highly specific, sensitivity reported from about 70% to 100% in recent series; confirmed in surgical series.
- **Limitations: up to 7% of adrenal metastases measure <10 HU** (European Society of Endocrinology–European Network for the Study of Adrenal Tumors, ESE–ENSAT), and **rare low-density pheochromocytomas** exist — metabolic work-up is mandatory for all lesions.
- **Measure density yourself** on digital images if the report omits it — avoids unnecessary further imaging.

Magnetic resonance imaging

- **Opposed-phase chemical-shift imaging:** fat–water signal interference; **signal loss on out-of-phase versus in-phase sequences = intracellular lipid = adenoma**. Usually assessed qualitatively (quantitative methods use the spleen as reference).
- **Test characteristics broadly equivalent to unenhanced CT;** possible advantage for lipid-poor adenomas is disputed. **CT washout is more sensitive than chemical-shift MRI.**

Contrast Enhancement

- **Post-contrast attenuation overlaps heavily** between adenomas and non-adenomas — enhanced phases are rarely useful alone, except for lesions that **do not enhance at all** (haemorrhage, cyst fluid).

- **Histogram analysis** of a region of interest on enhanced CT: if **>10% of pixels are <0 HU**, the lesion is probably an adenoma. Combination with fluorine-18 fluorodeoxyglucose positron emission tomography (¹⁸F-FDG PET) CT is promising.

Washout Studies

- **Lipid-poor adenomas wash out contrast as rapidly as lipid-rich ones** — the basis of the washout study, now routine for lipid-poor lesions; **highly sensitive and specific**.

ABSOLUTE VERSUS RELATIVE PERCENTAGE WASHOUT (15-MINUTE DELAY)

	Absolute percentage washout (APW)	Relative percentage washout (RPW)
Phases needed	Unenhanced, enhanced (60 s), delayed (15 min)	Enhanced and delayed only
Formula	$(\text{Enhanced} - \text{delayed}) \div (\text{enhanced} - \text{unenhanced}) \times 100$	$(\text{Enhanced} - \text{delayed}) \div \text{enhanced} \times 100$
Adenoma if	>60%	>40%
Below threshold	Adenoma very unlikely	

- **Worked example (Fig. 146.7):** unenhanced 15 HU, enhanced 55 HU, delayed 27 HU → $\text{APW} = 28/40 = 70\%$; $\text{RPW} = 28/55 \approx 51\%$ — both diagnostic of a lipid-poor adenoma.
- **Pitfalls: RCC and hepatocellular carcinoma metastases can wash out like lipid-poor adenomas.** Adrenocortical carcinoma (ACC) data are limited (usually large at presentation); ACC classically fails to wash out, with reported exceptions.
- **10-minute protocols:** RPW >50% was originally required (some say >40% suffices), but experts still recommend **15-minute delays**.
- **Order the protocol correctly and speak to the radiologist: unenhanced 5-mm adrenal images → enhanced (1 minute post-bolus) → 15-minute delayed.** A poorly performed test generates unnecessary, anxiety-provoking follow-up.
- **Gadolinium MRI washout is not used clinically** — gadolinium signal is less dose dependent than iodinated CT enhancement.

Fig. 146.8 — imaging algorithm (sequence)

- Adrenal mass → **<10 HU on CT or signal drop-out on chemical-shift MRI → adenoma**; **>10 HU or no drop-out → CT washout study (15-minute delay)** → $\text{APW} >60\%$ or $\text{RPW} >40\%$ = adenoma; below these = adenoma very unlikely.

Size

- **Larger masses are more likely malignant** (Fig. 146.9: benign lesions predominate below 4 cm; malignancy rises steeply above 6 cm).

RISK OF MALIGNANCY BY INCIDENTALOMA SIZE

Size	Malignancy (recent large series)	Other estimates in the chapter
<4 cm	0.1%	Low potential ($\approx 2\%$ carcinoma); 4–5% carcinoma in surgical series; observe if non-functional
4–6 cm	2.4%	$\approx 6\%$; >4 cm $\approx 10\%$
>6 cm	19.5%	$\approx 25\%$

- Estimates vary with **denominator** — unselected incidentaloma cohorts (lower) versus resected surgical series (higher).
- **4 cm is the consensus threshold to consider resection** in fit patients: a 4-cm cut-off gives high sensitivity but poor specificity for malignancy; 6 cm trades sensitivity for specificity.

- **Radiological size underestimates pathological size.**
- **Benign imaging overrides size:** in the largest series, **no incidentaloma <10 HU was malignant regardless of size**; in surgical series, all homogeneous adenoma-like lesions were benign whatever their size.
- **Age modifies the threshold:** adenoma incidence rises with age, so **small lesions in the young demand more caution**, while lesions >4 cm in older comorbid patients may be better observed. Borders, lipid content and enhancement are weighed with size; there is no level 1 evidence.

Growth

- **Follow growth kinetics of unresected lesions — but most resected for growth prove benign.** In a prospective cohort, a small minority grew ≥ 5 mm or ≥ 1 cm, and all resected for growth were benign.
- In a literature review, about **9% grew >1 cm over 3 years**; the rate of **malignant transformation ≈ 1 in 1000**.
- **ACC grows on average ≈ 1.9 cm per year.**

SURVEILLANCE AND GROWTH CRITERIA: CLASSIC VERSUS ESE-ENSAT

	Classic recommendation	ESE-ENSAT guideline
Re-imaging	6, 12 and (if possible) 24 months ; suspicious lesions earlier/more often	None for metabolically silent masses <4 cm with benign imaging ; others at 6–12 months
Growth trigger	≥ 1 cm growth (arbitrary) → consider resection; counsel that risk is low if kinetics flat	>20% increase in diameter and ≥ 5 mm = clinically concerning

KEY POINTS: SIZE AND GROWTH

- **Larger tumours are more likely malignant**, yet imaging features remain informative in large masses; in healthy, especially young, patients, **masses ≥ 4 cm should be resected or watched very closely**.
- **Follow growth kinetics** in lipid-poor masses or those >4 cm that are not resected.

Other Imaging Characteristics

- **Large heterogeneous masses invading surrounding structures suggest carcinoma**; **irregular margins and an enhancing rim** are quite specific for malignancy. Cysts and acute/subacute haemorrhage have characteristic appearances.

Macroscopic fat

- **Macroscopic fat almost always means myelolipoma** and is treated by most guidelines as **definitively benign** (reported 100% sensitive for myelolipoma).
- **No agreed definition:** areas < -30 HU proposed as fat density; series have used **-130 to -23 HU**. Also easily seen on MRI.
- **Fat proportion discriminates:** isolated myelolipomas are usually **>50% fat**; lesions with **<10% fat** were mixed myelolipoma with other pathology (hyperplasia, functioning or non-functioning adenoma). Rare fat-containing ACCs were symptomatic with **<5% fat**; “myelolipomatous degeneration” of adenomas had **<20% fat**. Teratoma, metastasis, lipoma and angiomyolipoma may contain fat (the latter two need not be distinguished clinically).
- **Distinguish from retroperitoneal liposarcoma:** aggressive, irregular borders, infiltrative. Some experts biopsy when in doubt.

Calcifications

- Some guidelines treat calcification as definitively benign, but it occurs in **up to 30% of ACCs** — the authors reject that confidence.
- **Fine punctate calcification suggests ganglioneuroma** but is not pathognomonic.

Other Imaging Modalities

Ultrasonography

- **Suboptimal for detection and characterisation**, yet the commonest source of incidentalomas where it is first-line imaging.
- **Right-sided lesions detected more often**: the IVC and liver give a window on the right; the left adrenal is hidden or mistaken for spleen, pancreas, para-aortic nodes or stomach.

Functional imaging

- **Rarely needed**; useful when the **origin is unclear** or **malignancy is suspected but unconfirmed**.
- **^{18}F -FDG PET**: high sensitivity and specificity for malignant lesions (ACC, metastases) but reserved for **inconclusive CT/clinical data**; CT washout may be more helpful in patients with known cancer. Caution again with **RCC and hepatocellular carcinoma** metastases.
- **Cortex-specific tracers**: **NP59** (radiolabelled cholesterol analogue) is taken up by adrenocortical cells — separates adenoma from metastasis; planar imaging is limited for small lesions, improved by single-photon emission CT (SPECT-CT). **^{11}C -metomidate** and **^{123}I -iodometomidate** (inhibitors of 11β -hydroxylase and aldosterone synthase) identify cortical origin but **cannot distinguish adenoma from ACC**, and ^{11}C has a short half-life.

Functional imaging in pheochromocytoma

FUNCTIONAL IMAGING FOR PHAEOCHROMOCYTOMA AND PARAGANGLIOMA

Modality	Role	Strength	Weakness
^{18}F-FDG PET	Gold standard for definitive staging (NIH data)	Superior to CT, MRI and MIBG, especially for metastases; exquisitely sensitive in SDHB and VHL ; avidity independent of biochemical phenotype	Poor in MEN-2 (<50% sensitivity) — far inferior to ^{123}I -MIBG there
^{68}Ga-DOTATATE PET/CT	Replacing FDG PET at some centres	Superior signal-to-noise; similar test characteristics	—
^{123}I-MIBG (norepinephrine analogue; preferred to ^{131}I)	Extra-adrenal, metastatic, recurrent disease; biochemistry positive but imaging negative	High sensitivity, superb specificity; best in MEN-2	Cold scans in VHL and SDHB ; aggressive metastases lose uptake

- Abbreviations: National Institutes of Health (NIH); succinate dehydrogenase subunit B (SDHB); von Hippel-Lindau (VHL); multiple endocrine neoplasia type 2 (MEN-2).
- **Solitary adrenal mass with positive biochemistry**: no consensus on routine MIBG — advocates cite confirmation and exclusion of missed metastases; others find it does not change management. **For large (>5 cm) tumours, MIBG or FDG PET is prudent** to exclude metastases before surgery.
- **Isolated norepinephrine/normetanephrine excess** (no epinephrine/metanephrine) → MIBG may reveal that the functioning lesion is **extra-adrenal**.
- Radiolabelled somatostatin analogues and **^{18}F -dopamine PET** are alternatives where MIBG fails; most have limited use.
- **Trap**: FDG PET is best in every genotype except MEN-2, where MIBG wins.

Figure legend — Fig. 146.10

- T2-weighted MRI of a right pheochromocytoma; MIBG shows **normal uptake in liver, salivary glands, thyroid and bladder**.

KEY POINTS: IMAGING

- **<10 HU on unenhanced CT** = high intracellular lipid — extremely specific for adenoma; **MRI is similar**.
- **CT washout is the gold standard for adrenal imaging**: it separates even lipid-poor adenomas (about 30% of adenomas) from other lesions; gadolinium MRI washout is not used.
- **RCC metastases may mimic lipid-poor adenomas**.

Concept — Adrenal imaging is a lipid assay followed by a perfusion assay: unenhanced density detects the fat that defines adenoma, and washout rescues the lipid-poor adenoma. Everything that defeats the method — RCC and hepatocellular metastases, low-density pheochromocytoma — is why imaging never replaces biochemistry.

Assessing for Hormonal Hypersecretion

Overview

- All adrenal lesions >1 cm need testing for hormone excess: cortisol and catecholamines in every case; aldosterone if hypertension or unexplained hypokalaemia.
- Rationale: >10% of incidentalomas are metabolically active (Table 146.4). Yet an estimated 75–80% of adrenal masses are inadequately evaluated.
- **Myelolipoma**: pure myelolipomas are never active, but the authors still advise hormonal work-up.
- **Catecholamine testing in <10 HU lesions**: observational data and a meta-analysis found **≥10 HU detected 100% of pheochromocytomas**, so testing may be unnecessary for <10 HU asymptomatic lesions — now incorporated in **Canadian Urological Association (CUA)** guidance.
- **Never biopsy, sample (AVS) or excise any adrenal lesion before excluding catecholamine excess** — manipulation can provoke a fatal catecholamine surge.

TABLE 146.4 CHARACTERISTICS OF INCIDENTAL ADRENAL MASSES IN A SYSTEMATIC REVIEW OF PUBLISHED SERIES (≥20 PATIENTS)

Lesion	% of all incidentalomas
Metabolically active	11.2
— Cortisol-producing adenoma	5.3
— Pheochromocytoma	5.1
— Aldosterone-producing adenoma	1.0
Malignant	7.2
— Adrenocortical carcinoma	4.7
— Metastasis	2.5
Total potentially surgical	18.4

Data from Young WF Jr, 2000 and 2007.

- **Pattern: nearly one in five incidentalomas is potentially surgical**; cortisol and pheochromocytoma dominate function, ACC dominates malignancy. **Trap**: aldosteronoma is the rarest functional lesion in incidentaloma series, though primary aldosteronism is common among hypertensives.

Testing for Aldosterone Hypersecretion

- **Only about 1% of adenomas cause Conn syndrome**, but up to ~5% of newly hypertensive patients may harbour an aldosterone-secreting adenoma.
- **2016 Endocrine Society guideline** (with nine other organisations) on primary aldosteronism (PA) — no competing guideline; the chapter follows it.

Indications for screening

- **Adrenal lesion plus hypertension or unexplained hypokalaemia.** Accurate diagnosis and subtyping matter for target-organ risk, drug selection and **potential surgical cure.**

BOX 146.2 INDICATIONS FOR PRIMARY ALDOSTERONE SCREENING

Category	Indication
Severity/ resistance	Resistant hypertension (≥ 3 agents including a diuretic); controlled on ≥ 4 agents; $\geq 150/\geq 100$ mmHg on 3 separate days
Biochemistry	Hypertension with hypokalaemia ; unexplained hypokalaemia (spontaneous or diuretic induced)
Associations	Adrenal incidentaloma with hypertension ; hypertension with sleep apnoea; target-organ damage disproportionate to hypertension
Age/family	Hypertension onset <20 years or stroke <50 years ; family history of PA
Work-up context	Whenever other secondary causes (phaeochromocytoma, renovascular) are considered

Modified from Funder JW et al., 2016.

Screening test: aldosterone-to-renin ratio

- **Screening test of choice: morning plasma aldosterone concentration (PAC, ng/dL) $\div$ plasma renin activity (PRA, ng/mL/h) = aldosterone-to-renin ratio (ARR)**, expressed as a unitless number; direct renin concentration (DRC) is an alternative.
- **Sample 8–10 am, preferably seated** — the same window as the morning cortisol of the dexamethasone test.
- **Floor the renin at 0.2 ng/mL/h** to avoid falsely high ratios; very low renin alone inflates the ARR, so a **high ARR with low PAC (especially <15 ng/dL) suggests a false positive** — but not absolutely, as some proven aldosteronomas have PAC <15 or even <10 ng/dL.
- **Hypokalaemia is an unreliable screen:** fewer than 40% of patients with PA are hypokalaemic.

ARR THRESHOLDS PROPOSED

Source	ARR	PAC
Range in the literature	20–40 (20 most common ; as low as 15 in some settings)	—
NIH Consensus Statement 2002	>30	>20 ng/dL
Other institutions	>20	>15 ng/dL

- Thresholds vary with laboratory; **lower cut-offs raise false positives**, acceptable because **every positive screen must be confirmed**. Sensitivity and specificity vary widely between series.

*Preparation and drug effects (Fig. 146.11)***DRUG EFFECTS ON ARR SCREENING**

Agent	Effect	Action
Mineralocorticoid receptor antagonists (spironolactone), potassium-sparing diuretics (amiloride)	Greatly alter RAAS	Stop before testing — at least 4 weeks (Endocrine Society 2016) or about 6 weeks (other experts; Figs 146.11, 146.13); substitute $\alpha 1$-blockers or long-acting calcium channel blockers if needed
β-blockers	Aldosterone unchanged, renin $\downarrow \rightarrow$ ARR $\uparrow$ (false positive)	May continue — aldosterone stays normal; some stop them (for several weeks)
ACE inhibitors, angiotensin receptor blockers (ARBs)	Aldosterone $\downarrow$, renin $\uparrow \rightarrow$ ARR $\downarrow$ (small chance of false negative)	No need to stop: autonomous secretion is RAAS independent, and an undetectable renin despite these drugs triggers confirmatory testing even with a normal ARR

Agent	Effect	Action
Calcium channel blockers	Neutral	No need to stop

- Also: **correct hypokalaemia** (may cause false results), **liberalise sodium**, avoid **liquorice** (including chewing tobacco). Historical advice to stop all antihypertensives is no longer necessary. ACE, angiotensin-converting enzyme.
- **Trap:** β -blockers cause false positives; ACE inhibitors and ARBs cause false negatives — opposite directions.

Confirmatory testing

- **Mandatory after a positive screen** — only 50–87% of positive screens are confirmed. **Refer to an experienced endocrinologist** at this point.
- **Omit confirmation** when there is **spontaneous hypokalaemia + undetectable renin + PAC >20 ng/dL** (high positive predictive value).
- Most tests exploit **failure of aldosterone to suppress with sodium loading**; no test is superior, so choice is by patient and physician. **Monitor blood pressure**; correct hypokalaemia and stop mineralocorticoid receptor antagonists first.

CONFIRMATORY TESTS FOR PRIMARY ALDOSTERONISM

Test	Protocol	Positive (PA)	Comment
Oral sodium loading	High-sodium diet 3 days (≥ 12.8 g sodium/day) → 24-h urine aldosterone, sodium, creatinine; urine sodium must be >200 mmol	24-h urine aldosterone >12 $\mu\text{g/day}$	Preferred with IV saline
Intravenous saline infusion	2 L 0.9% saline over 4 h , recumbent, morning after overnight fast	PAC >5 ng/dL; >10 ng/dL suggests adenoma	Spares days of loading
Fludrocortisone suppression	Fludrocortisone 0.1 mg 6-hourly + NaCl 2 g 8-hourly $\times$ 4 days; upright PAC	PAC fails to suppress <6 ng/dL	Former gold standard; abandoned for severe hypertension and hypokalaemia risk
Captopril suppression	Captopril 25–50 mg, recumbent	PAC stays >15 ng/dL	Needs validation; for patients in whom sodium loading is contraindicated (cardiac, renal disease)

- Furosemide-upright and ACTH stimulation tests exist but are not widely used. **Trap:** sodium loading is the principle; captopril is the escape route when salt cannot be given.

Subtype differentiation

- **Essential because only some subtypes are surgically curable** (Box 146.3). **Non-surgical candidates skip subtyping** and start medical therapy (Fig. 146.12).
- **Genetic testing not for all**; consider with **family history of PA, onset <20 years, or family history of early stroke**.
- **Adrenal CT in all potential surgical candidates**. Aldosterone-producing adenoma: **unilateral, <10 HU, non-enhancing, average 1.6–1.8 cm, normal contralateral gland** — but **about 20% are <1 cm** (Fig. 146.14: a typical ~ 1 -cm aldosteronoma), so **lateralisation is never based on CT alone**. Idiopathic hyperplasia: multiple nodules, bilateral limb enlargement, or normal glands. **MRI adds cost without advantage**.
- **AVS establishes lateralisation** in surgical candidates: CT-based decisions would have wrongly excluded 22% and wrongly recommended adrenalectomy in 25%. When done correctly, **AVS sensitivity 95%, specificity 100%**.

- **AVS exceptions:** age <40 with a clear unilateral adenoma (>1 cm) and normal contralateral gland (incidental adenomas rare in the young), and suspected ACC.
- **SPARTACUS randomised trial (CT- versus AVS-guided treatment):** no difference in antihypertensive intensity at 1 year, challenging routine AVS; CT and AVS lateralisation were **discordant in half**; persistent biochemical aldosteronism after surgery was more frequent with CT guidance (20% vs 10%).

BOX 146.3 SUBTYPES OF PRIMARY ALDOSTERONISM

Surgically correctable	Not correctable by surgery
Aldosterone-producing adenoma	Bilateral adrenal hyperplasia
Primary unilateral adrenal hyperplasia	Familial hyperaldosteronism type I
Ovarian aldosterone-secreting tumour	Familial hyperaldosteronism type II
Aldosterone-producing carcinoma	Familial hyperaldosteronism type III

Fig. 146.12 and Fig. 146.13 — algorithms (sequence)

- **Adrenal mass** → hypertension? No → no aldosterone testing; yes → morning ARR and aldosterone (stop potassium-sparing diuretics 6 weeks, consider stopping β -blockers) → abnormal → confirmatory testing with endocrinology $\pm$ AVS; normal → PA unlikely.
- **Indication for screening** → PAC/PRA → positive → confirm → PA → family history or onset <20 → screen for familial hyperaldosteronism type I → surgical candidate? No → medical management; yes → adrenal CT → age <40 with unilateral nodule >1 cm → **adrenalectomy**; otherwise (normal glands, bilateral enlargement, micronodularity, or age $\geq$ 40) → AVS → lateralised → adrenalectomy; not lateralised → medical management.

Adrenal vein sampling technique (Fig. 146.15)

- **Preparation:** 1 h recumbency, correct hypokalaemia, withdraw interfering antihypertensives; morning, after overnight fast.
- Femoral venous access → catheters in **right adrenal vein, left adrenal vein and IVC** → measure **aldosterone and cortisol** in each.
- **Selectivity:** adrenal vein cortisol $\div$ IVC cortisol must exceed **1.1:1 to 5:1** (higher with cosyntropin). Below cut-off on either side = **non-selective** → **discard** (consider repeat, with ACTH stimulation if not used).
- **Lateralisation:** $(A_{\text{dominant}}/C_{\text{dominant}}) \div (A_{\text{non-dominant}}/C_{\text{non-dominant}}) > 2:1$ to **4:1** (higher with cosyntropin) → consider ipsilateral adrenalectomy; otherwise medical management.

COSYNTROPIN STIMULATION DURING AVS

For	Against
Minimises stress-related aldosterone fluctuation; maximises cortisol gradient (easier confirmation of selectivity); maximises aldosteronoma secretion	Small risk of wrong-side lateralisation ; moderate risk of misclassifying adenoma as idiopathic hyperplasia

When AVS is inconclusive

ADJUNCTS FOR SUBTYPE DIFFERENTIATION

Test	Principle	Adenoma pattern	Limitation
NP59 scintigraphy (^{131}I -6 β -iodomethyl-norcholesterol, cholesterol analogue)	Only modality giving functional and anatomical data; Lugol iodine thyroid protection; dexamethasone 1 mg 6-hourly $\times$ 7 days; image days 4–10	Unilateral uptake <5 days ; bilateral = hyperplasia	Poor for lesions <1.5 cm; no longer available in the USA ; SPECT-CT improves sensitivity

Test	Principle	Adenoma pattern	Limitation
Posture stimulation	Overnight recumbent vs 4 h upright PAC; upright angiotensin ↑	Adenoma and familial hyperaldosteronism I: no rise ; hyperplasia: ≥33% rise ; normal: 2–4-fold	No side information; 30–50% of adenomas rise, 20% of hyperplasia do not
18-hydroxycorticosterone (aldosterone precursor)	Serum level	>100 ng/dL suggests adenoma	Accuracy <80%; no side information
¹¹C-metomidate PET-CT	Cortical enzyme inhibitor tracer	Detects aldosteronoma with moderate sensitivity, good specificity	Needs validation and availability

Testing for Hypercortisolism

- **All lesions >1 cm; 5–8% of incidentalomas** secrete excess glucocorticoid.
- **Overnight low-dose dexamethasone suppression test (LD-DST) is the favoured screen** for adrenal masses. Late-night salivary cortisol (LNSC), 24-hour urinary free cortisol (UFC) and the 48-hour 2-mg test are valid for Cushing syndrome generally (2008 Endocrine Society) but **secondary for adrenal masses**.
- **Exclude exogenous steroids**, including creams and nasal sprays, before testing. Before resection for clinically significant hypercortisolism, **confirm suppressed/low morning ACTH**.
- Other conditions activate the HPA axis and mimic Cushing syndrome (Box 146.4) — **endocrine expertise is advised**.

SPECIAL POPULATIONS: CHOICE OF CORTISOL SCREEN

Population	Test	Reason
Pregnancy	UFC	Avoids dexamethasone; cut-offs raised 2–3-fold in 2nd–3rd trimesters
Epilepsy	LNSC	Antiepileptics accelerate dexamethasone clearance
Renal failure	LD-DST	Urinary cortisol falls when creatinine clearance <60 mL/min; LNSC norms not established

Terminology

CORTISOL-EXCESS TERMINOLOGY

Term	Meaning
Hypercortisolism / hypercortisolaemia / cortisol excess	Excess circulating cortisol
Cushing syndrome	Hypercortisolism with observable clinical signs (skin fragility, proximal weakness, fat redistribution)
Cushing disease	Cushing syndrome from a pituitary ACTH-secreting adenoma
Autonomous cortisol secretion/excess	Neoplastic cortisol excess without ACTH drive — a biochemical diagnosis
Mild autonomous cortisol secretion (MACS)	Autonomous secretion without clinical signs ; replaces “subclinical Cushing syndrome” because it is not truly subclinical — ↑ cardiovascular mortality, bone disease, diabetes, hypertension

SCREENING TESTS FOR CORTISOL EXCESS

Test	Method	Threshold	Pitfalls
Overnight 1-mg LD-DST (preferred)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	>5.0 µg/dL (140 nmol/L): >95% specific; ≤1.8 µg/dL (50 nmol/L): >90% sensitive — rules out; 1.8–5.0 = “possible autonomous cortisol secretion”	Oral contraceptives → up to 50% false positives (↑ cortisol-binding globulin); CYP3A4 inducers/inhibitors (Box 146.5); not affected by GFR

Test	Method	Threshold	Pitfalls
Late-night salivary cortisol	Saliva 11 pm–midnight or bedtime; repeat at least twice	≤145 ng/dL by liquid chromatography–tandem mass spectrometry (LC-MS/MS)	False positives in depression, altered sleep, chronic illness; no tobacco that day; no food, drink, tooth-brushing or activity 30 min before; unvalidated in chronic kidney disease ≥stage III; weaker evidence base; doubtful for MACS
24-h urinary free cortisol	Integrated free cortisol; two collections; check creatinine for completeness	Normal ≈ 50–100 µg/24 h (assay dependent)	Insensitive for subtle adrenal cortisol excess; unreliable if GFR <60 mL/min ; unaffected by binding globulin; test of choice in pregnancy

- Despite the name, **1 mg dexamethasone is 3–4 times physiological glucocorticoid**, and the **assay does not detect dexamethasone** — so normal subjects have low cortisol. At the 5.0 threshold, up to 18% of Cushing syndrome gives **false negatives**, hence the 1.8 rule-out cut-off (specificity about 80%).
- If LD-DST morning cortisol is **>1.8 µg/dL**: repeat metabolic testing and draw basal morning ACTH. A 48-hour low-dose test may improve accuracy.
- **Trap**: “low-dose” is supraphysiological; and the **UFC is displaced as first-line screen** by LD-DST or LNSC for incidentalomas (2008 guidance).

BOX 146.4 CAUSES OF HYPERCORTISOLISM IN THE ABSENCE OF CUSHING SYNDROME

Some cushingoid features possible	Cushingoid features unlikely
Pregnancy; depression; alcohol dependence; glucocorticoid resistance; morbid obesity; poorly controlled diabetes	Physical stress (hospitalisation, surgery, pain); malnutrition, anorexia nervosa; intense chronic exercise; hypothalamic amenorrhoea; cortisol-binding globulin excess (serum, not urine, cortisol ↑)

Modified from Nieman LK et al., 2008. Cushing syndrome may rarely coexist; test if suspicion is high, especially in the first group.

BOX 146.5 PHARMACEUTICALS THAT AFFECT TESTING FOR CUSHING SYNDROME

Mechanism	Drugs	Effect on test
CYP3A4 induction — accelerate dexamethasone metabolism	Phenobarbital, phenytoin, carbamazepine, primidone, rifampicin, rifapentine, ethosuximide, pioglitazone	False-positive DST
CYP3A4 inhibition — impair dexamethasone metabolism	Aprepitant/fosaprepitant, itraconazole, ritonavir, fluoxetine, diltiazem, cimetidine	False-negative DST
↑ Cortisol-binding globulin	Oestrogens, mitotane	Falsely high serum cortisol
↑ Urine free cortisol	Carbamazepine; fenofibrate (HPLC); some synthetic glucocorticoids (immunoassay); 11β-hydroxysteroid dehydrogenase type 2 inhibitors (liquorice, carbenoxolone)	Falsely high UFC

From Nieman LK et al., 2008; not a complete list.

Confirmatory testing

- **Clinical signs plus abnormal screen**: confirmation probably unnecessary. **Abnormal test with low suspicion, or normal tests with high suspicion** → endocrinologist.

Identifying the cause of hypercortisolism (sequence)

- **Step 1 — exclude exogenous glucocorticoid**: the **commonest cause in the Western world**; oral, topical or inhaled, even low dose; ask about herbal remedies, nasal sprays and surreptitious use (e.g., performance enhancement).

- **Step 2 — serum ACTH:**
 - **Low ACTH → ACTH-independent:** adrenal adenoma (autonomous secretion) or PPNAD; also exogenous steroid. **Normal adrenals on imaging → suspect exogenous steroid, rarely PPNAD.** Up to 10% of adrenal Cushing syndrome is bilateral — AVS may help.
 - **High ACTH → ACTH-dependent** (pituitary adenoma or ectopic source) — outside urology; refer.
- **Step 3 — pituitary versus ectopic:** difficult because both sources are hard to image and incidental pituitary, lung and pancreatic findings are common (10% of the population have a pituitary imaging abnormality; half of Cushing disease has a normal pituitary MRI). **Gold standard: inferior petrosal sinus ACTH sampling after CRH stimulation** — central-to-peripheral gradient = Cushing disease; no gradient = ectopic.
- **High-dose dexamethasone suppression is no longer routine:** some ectopic tumours carry glucocorticoid receptors and also suppress.

Fig. 146.16 — cortisol algorithm (sequence)

- Adrenal mass → exclude exogenous glucocorticoid → **one of:** overnight 1-mg DST, LNSC (×2) or 24-h UFC (×2) → normal = Cushing syndrome unlikely; any abnormal → endocrinologist for confirmation (repeat abnormal LNSC/UFC).

Testing for Adrenal Sex Steroid Hypersecretion

- **Exceedingly rare** in incidentalomas. **Commonest sex-steroid-secreting mass: ACC co-secreting cortisol.** Pure androgen-secreting tumours (testosterone, 17-ketosteroids) occur mainly in women; about half are benign.
- **No routine testing; test when ACC is suspected:** DHEA-S, androstenedione, 17-hydroxyprogesterone; **testosterone in women; oestradiol in men and postmenopausal women** — potential ACC tumour markers; diagnostic value for benign versus malignant not yet defined.
- Dexamethasone suppression to separate adrenal from ovarian androgen is unreliable — replaced by imaging.

Testing for Catecholamine Hypersecretion

- **Phaeochromocytoma in about 5% of incidentalomas → all patients, including those with suspected metastasis, need testing.**
- **Broad guideline consensus: plasma free metanephrines or urinary fractionated metanephrines;** performance is similar overall.
- **Liquid chromatography methods (electrochemical/fluorometric or tandem mass spectrometry) are preferred to immunoassays.**
- **Interference: paracetamol (acetaminophen), mesalazine, sulfasalazine** cross-react with electrochemical detection — **stop ≥5 days; tricyclic antidepressants and phenoxybenzamine** cause false positives — stop; **severe illness** causes false positives.
- **Urine catecholamines and VMA are abandoned** as first-line tests (lower accuracy).

TABLE 146.5 RELATIVE MERITS OF URINARY FRACTIONATED VERSUS PLASMA FREE METANEPHRINES

Issue	Urinary fractionated metanephrines	Plasma free metanephrines
Availability	Well established, widely available	Newer, limited availability
Analytical ease	High concentrations (200–2000 nmol) — easy	Low concentrations (0.1–0.5 nmol) — harder
Staff burden	Minimal	Blood collection needs staff time
Patient convenience	24-h collection inconvenient; incomplete collections unreliable	More convenient; collection and handling better regulated

Issue	Urinary fractionated metanephrines	Plasma free metanephrines
Diet/sympathetic influences	Hard to control	More easily controlled
Children	Needs age-appropriate reference ranges	Sampling stressful, but interpretation easier
Renal failure	May be inappropriate	Applicable

From Grossman A et al., 2006.

PLASMA FREE VERSUS 24-H URINARY FRACTIONATED METANEPHRINES: PROTOCOL AND THRESHOLDS

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Near-perfect sensitivity (sporadic and familial); specificity lower in sporadic cases	By tandem mass spectrometry: sensitivity >97%, specificity ≈91%; some pair it with urinary catecholamines as best first line
Conditions	Supine, after 20 min rest (seated sampling raises normal values and blurs separation); fast from midnight; no caffeine ; continue antihypertensives (stop β-blockers only on repeat)	Discard first morning void, collect all urine to next first morning void; check creatinine; normal renal function
Upper limit of normal	Normetanephrine 0.61 nmol/L (112 ng/L); metanephrine 0.31 nmol/L (61 ng/L)	Positive: metanephrine >1531, normetanephrine >4001, total >1563 nmol/day
Highly indicative	Normetanephrine >2.2 nmol/L (400 ng/L) and/or metanephrine >1.2 nmol/L (236 ng/L)	—

- **Lesser elevations → repeat:** repeat plasma test under ideal conditions (supine 20 min, off β-blockers), urinary fractionated metanephrines, **chromogranin A**, or **clonidine suppression** — and involve an experienced endocrinologist.

Fig. 146.21 — catecholamine algorithm (sequence)

- Adrenal mass → stop tricyclics and phenoxybenzamine → **plasma free metanephrines** (stop paracetamol 5 days; supine) or **24-h urinary fractionated metanephrines** (verify renal function) → normal = pheochromocytoma extremely unlikely; any abnormal → endocrinologist for confirmation.

KEY POINT: ASSESSMENT OF FUNCTION OF ADRENAL MASSES

- **All adrenal incidentalomas demand a metabolic evaluation.**

Concept — Every hormonal screen is a suppression-or-rhythm test for autonomy: aldosterone that ignores a suppressed renin or a salt load, cortisol that ignores dexamethasone or the night, catecholamine metabolites produced continuously inside the tumour. Each false result comes from something that shifts the regulator or the assay — drugs, binding globulins, posture, renal function.

Biopsy of Adrenal Masses

Overview

- **Limited role:** (1) imaging with clinical data diagnoses well; (2) **histology cannot reliably separate adenoma from carcinoma**; (3) biopsy carries risk.
- **ESE-ENSAT: no biopsy without a history of malignancy.** Main role: **confirm suspected metastasis** (rarely necessary); tissue in **metastatic ACC before systemic therapy**.

Test Characteristics of Biopsy

- **Without known malignancy:** solitary adrenal metastasis as first presentation is extremely rare; **sensitivity for malignancy at best 81–84%, for ACC at best 70%**; specificity ≈99%. **18–19-gauge needles** improve yield without more complications; adequacy about 88% (67–100%).

- **With known extra-adrenal malignancy: >50% of new adrenal lesions are metastases;** biopsy accurately separates metastasis from benign lesion, with excellent **negative predictive value** once non-diagnostic samples are excluded. Positive yield highest with **lung cancer (>90%) and RCC (80%)**.

Complications of Biopsy

- **2.8–12%; bleeding commonest;** also pneumothorax, haemothorax, abscess, bacteraemia, pancreatitis. Biopsy (especially with haemorrhage) may **complicate or prevent laparoscopic resection**.
- **Needle-tract seeding in ACC** — some advise avoiding biopsy if ACC suspected (only two cases reported).
- **Exclude pheochromocytoma biochemically first** — biopsy can precipitate a life-threatening hypertensive crisis (some inadvertent biopsies were uneventful, but due diligence is mandatory).

Clinical Usefulness of Biopsy

- Community biopsies before tertiary referral **never changed management** in one review. Radiology reports calling lesions “amenable to biopsy” should be **strongly discouraged**.
- **Biopsy only when imaging limits are reached and the result will change management** — chiefly a **known primary cancer possibly recurring in the adrenal**.

Fig. 146.17 — biopsy decision (sequence)

- Can imaging characterise the mass? Yes → no biopsy. Metabolically active? Yes → no biopsy. ACC strongly suspected? Yes → no biopsy. Will biopsy change management, and does benefit justify risk (including harder laparoscopic surgery)? Yes → **biopsy, with pheochromocytoma excluded**.

KEY POINTS: BIOPSY OF ADRENAL MASSES

- **Favourable test characteristics for benign versus metastatic disease.**
- **Always exclude pheochromocytoma before biopsy.**

Concept — *Biopsy answers only one question well — is this a metastasis? — because cytology cannot separate adenoma from carcinoma. It is therefore justified only when a known cancer may have recurred and the answer changes treatment.*

ADRENAL NEOPLASIA

Adrenal Incidentaloma

Definition

- **Adrenal mass >1 cm found on cross-sectional imaging performed for an unrelated reason.** Strictly excludes masses found during **cancer staging** or in patients later found to have **adrenal-related symptoms**.
- Work-up aims to **reclassify the incidentaloma into a specific neoplasm**, which then dictates management.

Epidemiology

- **About 5% on imaging**, matching autopsy data; rises with age from **<0.5% in the 20s to up to 7% at ≥70 years**.
- The retroperitoneum allows silent growth beyond 20 cm, yet **median size at presentation ≈3 cm**.
- **“Incidental” does not mean “insignificant”**: nearly **20% are potentially surgical** (Table 146.4).

Adrenal Adenoma

Overview and Epidemiology

- **Commonest adrenal neoplasm**, cortical in origin; about 6% at autopsy; **>85% of incidentally imaged adrenal neoplasms are adenomas**; prevalence rises with age.
- Task: **separate the non-functional benign adenoma from functional or malignant lesions**.

Pathophysiology and Clinical Characteristics

- **Yellow from abundant lipid**; histologically may resemble carcinoma — distinguished by the **Weiss criteria** and immunohistochemistry (Box 146.6).
- **Benign by definition; most metabolically silent**: in pooled series ≈90% of incidentalomas were benign adenomas, of which ≈93% were inactive, ≈6% cortisol-secreting and ≈1% aldosterone-secreting.
- **Testosterone-secreting adenomas** are exceedingly rare and may contain **Reinke crystals** (normally in Leydig cells).
- Associations of non-functional adenoma with insulin resistance and left ventricular dysfunction are **unconfirmed**.
- Most remain stable; about 9% grow >1 cm over 3 years.

Hormonally Inactive Adenoma

- **No immediate intervention**. Repeat testing and imaging are **controversial**, with society disagreement.

FOLLOW-UP OF HORMONALLY INACTIVE ADENOMA: GUIDELINE BODIES

Body	Repeat hormonal testing
European Society of Endocrinology	None unless new symptoms of hormone excess
Canadian Urological Association; American Association of Clinical Endocrinology; Korean Endocrine Society	Annual

- Evidence: risk of new clinically apparent hormone excess is **<1–1.7%** (aldosterone ≈0%, cortisol ≈1%, catecholamine ≈0.05%), possibly higher with bilateral adenomas.
- **Imaging**: in the largest series, **no lesion <10 HU became malignant** and only 0.2% of lesions <4 cm were malignant → **no follow-up imaging if both features present; probably none if either**. Indeterminate unresected lesions: re-image at **6, 12 and possibly 24 months**; suspicious lesions earlier, more often, or remove.

Aldosterone-Secreting Adenoma

Overview and epidemiology

- **Conn (1955)**: hypertensive, hypokalaemic 34-year-old woman cured by removal of a 4-cm unilateral tumour.
- **PA is the commonest secondary hypertension**: 3–13% of hypertensives in primary care, up to 30% at tertiary centres.

Pathophysiology

- **RAAS (Fig. 146.18)**: **renin release is rate limiting**, stimulated by low renal perfusion, renal sympathetic activity and low macula densa sodium → angiotensinogen → angiotensin I → (ACE) angiotensin II → vasoconstriction and aldosterone release (also potassium, ACTH).

PRIMARY VERSUS SECONDARY HYPERALDOSTERONISM

Primary	Secondary
Aldosterone autonomous → renin suppressed	Renin elevated → drives aldosterone
• No hypernatraemia: sodium carries water. Volume expansion is capped at about 1.5 kg by mineralocorticoid escape (pressure natriuresis, atrial natriuretic peptide, distal transporter changes). • Idiopathic (bilateral) hyperplasia and aldosterone-producing adenoma account for >95%; hyperplasia is diagnosed more often, but adenomas may be under-counted because lateralisation is not routinely performed.	

FIG. 146.19 SUBTYPES OF PRIMARY ALDOSTERONISM

Subtype	Proportion
Bilateral hyperplasia	60%
Adenoma	35%
Unilateral hyperplasia	2%
Carcinoma; ectopic tumour; familial hyperaldosteronism	<1% each

ALDOSTERONE-PRODUCING ADENOMA VERSUS IDIOPATHIC HYPERPLASIA

	Aldosterone-producing adenoma	Idiopathic (bilateral) hyperplasia
Hypertension	More profound	Less severe
Hypokalaemia	≈50%	≈17%
Morphology	Small unilateral nodule	Mostly micronodular (macronodular rare); cause unknown
Surgery	Curable by unilateral adrenalectomy	Unilateral adrenalectomy not therapeutic

- **Unilateral hyperplasia:** distinctly uncommon; curable once lateralisation is confirmed.
- **ACC with PA:** <1% of PA; 2.5–5% of ACC. **Ectopic aldosterone-secreting tumours** suggest accessory tissue or heterotopia.

FAMILIAL HYPERALDOSTERONISM (FH)

Type	Genetics	Phenotype	Test whom / treat
FH-I (glucocorticoid-remediable aldosteronism)	Autosomal dominant; chimeric CYP11B1 promoter fused to CYP11B2 coding sequence → aldosterone synthase under ACTH control, secreted on the ACTH circadian rhythm	Early-onset refractory hypertension; family history of early hypertension or stroke	Test with family history of PA or young stroke, or PA <20 years; treat with oral glucocorticoid (± mineralocorticoid receptor antagonist if uncontrolled or iatrogenic Cushing)
FH-II	Autosomal dominant, heterogeneous; 7p22 linkage in some	Hyperplasia or adenoma indistinguishable from sporadic PA	No specific testing recommendation
FH-III	KCNJ5 (potassium channel) mutation	Bilateral hyperplasia, refractory hypertension, severe hypokalaemia, hybrid steroids	Test with PA at a very young age

Clinical characteristics

- Usually diagnosed in the **3rd–6th decades** during work-up of refractory, **moderate-to-severe hypertension**; prevalence of PA rises with hypertension stage.
- **Most are normokalaemic (63–91%);** hypokalaemic symptoms (headache, polydipsia, polyuria, nocturia, palpitations, weakness) occur in a minority. **Serum sodium resembles other hypertension** because of escape.

- **Target-organ damage exceeds essential hypertension:** left ventricular diastolic dysfunction and hypertrophy, prolonged QT; about **4-fold stroke, 6.5-fold myocardial infarction, 12-fold atrial fibrillation**; more proteinuria, type 2 diabetes and metabolic syndrome. **Treatment induces left ventricular remodelling.**

Treatment and prognosis

- **Goal:** remove the source or block aldosterone at target organs, chosen by subtype and surgical fitness (Fig. 146.12).
- **Lateralised secretion → consider adrenalectomy**, usually **laparoscopic** (small tumours); **open if ACC suspected.**
- Outcomes: blood pressure improves in most, but only **37–52% need no antihypertensives**; **biochemical success ≈94%** — residual hypertension reflects other conditions.
- **Predictors of persistent hypertension:** age >50, male, high body mass index, preoperative ARR, ≥2 preoperative agents, first-degree relative with hypertension, long duration, cardiovascular remodelling, renal insufficiency.
- **Age <40 with unilateral adenoma >1 cm → adrenalectomy without AVS.**
- **Postoperatively:** check PAC and ARR for biochemical cure; **high-sodium diet and monitoring for hyperkalaemia for several weeks**; blood pressure continues to change for **up to 6 months.**

MINERALOCORTICOID RECEPTOR ANTAGONISTS IN PA

Drug	Start	Maximum	Profile
Spironolactone	25–50 mg/day	400 mg/day	Gynaecomastia, erectile dysfunction, menstrual disturbance
Eplerenone	25 mg/day	100 mg/day	More selective → fewer side effects

- **Agents of choice** for non-surgical subtypes and unfit patients; titrate to blood pressure, potassium and tolerability. Other antihypertensives are often still needed; add **weight loss, low-sodium diet and exercise.**
- **Summary:** PA carries increased cardiovascular morbidity; because subtypes dictate management, **subtype differentiation is critical before surgery.**

KEY POINTS: PRIMARY ALDOSTERONISM

- **Most patients are normokalaemic** in contemporary series.
- **End-organ damage exceeds that of essential hypertension.**
- **Accurate subtype differentiation** ensures appropriate treatment.

Concept — Primary aldosteronism is diagnosed by the renin it suppresses and treated by the side it comes from: a unilateral source is cured by adrenalectomy, a bilateral source only blocked by mineralocorticoid receptor antagonists. That is why lateralisation, not imaging, decides operation.

Cortisol-Secreting Adenoma

- **About 10% of Cushing syndrome**; usually a **dominant unilateral hyperplastic nodule**, sometimes multifocal bilateral hyperplasia. **<10% of adrenal masses are bilateral** and these are diagnostically challenging; radiologically invisible lesions may cause MACS.

Additional testing

- **Confirm hypercortisolism and its ACTH-independent cause before resection.**
- Screen for **vertebral fractures** (once at diagnosis — plain films, or existing CT for included levels), **type 2 diabetes** and **hypertension** (general-population guidelines for the latter two). No consensus on dual-energy X-ray absorptiometry despite common osteoporosis.

- **Hypogonadism is common in men** — test if symptomatic. **Urolithiasis in up to 50% of Cushing syndrome** — no screening recommendation.

Mild autonomous cortisol secretion

- **Adrenalectomy is controversial — individualise (ESE-ENSAT)**. Surgery may improve glucose control, hypertension and weight, but **progression to overt Cushing syndrome is extremely unlikely**.
- Opposing views: operate only when signs present (hypertension, obesity, glucose intolerance, osteopenia) versus operate on all to prevent sequelae.

Postoperative care

- **Postoperative adrenal insufficiency: ≈65% after unilateral adrenalectomy for MACS; ≈99.7% for overt Cushing syndrome** — autonomous cortisol **suppresses and atrophies the contralateral gland**.
- **Endocrine Society replacement: hydrocortisone 10–12 mg/m²/day in 2–3 doses**, first dose on waking.
- **Common perioperative protocol: hydrocortisone 100 mg IV on the day of surgery** → slow oral taper (cortisone acetate 25–37.5 mg/day or hydrocortisone 25–50 mg/day) once oral intake resumes.
- Deaths from hypocortisolism have been reported after surgery for Cushing syndrome, **none after MACS**.
- **Alternative: dexamethasone 0.5 mg 6-hourly from surgery** (not detected by the cortisol assay) → **early-morning cortisol <50 nmol/L confirms successful resection** → switch to hydrocortisone.
- **Taper: unstudied; one scheme checks morning cortisol 3-monthly** and starts ACTH stimulation tests once it exceeds **7.4 µg/dL**.

Prognosis

- **Median recovery of adrenal function: 6.5 months (MACS) vs 11 months (Cushing syndrome)**; a few (≈2% and ≈7%) never recover.
- Normalising cortisol **reduces but does not eradicate cardiovascular risk**; cushingoid features resolve in **7–9 months** in most, occasionally persisting for years.

Concept — A cortisol-secreting adenoma switches off the ACTH that sustains the opposite gland, so its removal leaves an atrophic contralateral adrenal. The degree of preoperative autonomy predicts how certain and how long the postoperative adrenal insufficiency will be.

Phaeochromocytoma

Overview and Epidemiology

- **Tumour of medullary catecholamine-producing cells; 1–2 per 100 000 per year**; causes only ≈0.5% of hypertension.
- ≈5% of **incidentalomas**; incidental discovery accounts for **10–25% of all diagnoses**.
- **Paragangliomas = extra-adrenal phaeochromocytomas** (10–25%; Fig. 146.20: adrenal 75–90%) arising from paraganglia along sympathetic and parasympathetic chains — head, neck, thorax, abdomen, pelvis including **bladder**. The **organ of Zuckerkandl** (aortic bifurcation to inferior mesenteric artery root) is a common site. Some reserve “paraganglioma” for head and neck tumours, which are usually non-functional.

Pathophysiology

Overview

- **Adrenal phaeochromocytomas possess PNMT** and can make epinephrine; tumours vary in enzyme content and self-metabolism, so **secretory profiles and clinical behaviour vary**.

EPINEPHRINE-DOMINANT VERSUS NOREPINEPHRINE-DOMINANT TUMOURS

Epinephrine-secreting (rare; adrenal or organ of Zuckerkandl)	Norepinephrine-secreting (e.g., VHL)
Syncope, hypotensive episodes — β_2 vasodilatation	Hypertension, sweating — α vasoconstriction

Hereditary phaeochromocytoma

- **Familial in about one-third; $\approx 24\%$ of apparently sporadic cases without family history carry a germline mutation.**
- Classic genes: **RET, VHL, NF1, SDHD, SDHB** — possibly converging on neuronal apoptosis downstream of nerve growth factor, yet with diverse phenotypes. **Fourteen susceptibility genes** reported since 1990.
- **VHL and SDHB matter most to urologists** — these patients are already under urological care for RCC risk.

VHL CLINICAL SUBTYPES

Type	Phaeochromocytoma	RCC
1	No family history of phaeochromocytoma	—
2A	Yes	No
2B	Yes	Yes
2C	Phaeochromocytoma only, no other VHL stigmata	No

- Type 2 disease is associated with **missense VHL mutations**; type 2C retains hypoxia-inducible factor (HIF) regulation, so tumorigenesis may be **HIF independent**.
- **VHL tumours make little epinephrine $\rightarrow$ metanephrine normal, normetanephrine raised.**
- **SDHB penetrance $\approx 9\text{--}20\%$ by age 50** after correction for ascertainment bias.

TABLE 146.6 HEREDITARY FORMS OF PHAEOCHROMOCYTOMA

Syndrome	Features	Phaeo risk	Gene; locus; inheritance; protein	Germline in apparently sporadic (%)	Malignant (%)
MEN-2A	Medullary thyroid cancer, hyperparathyroidism, cutaneous lichen amyloidosis	50%	RET; 10q11.2; AD; tyrosine kinase receptor	<5	3
MEN-2B	Medullary thyroid cancer, multiple neuromas, marfanoid habitus, rare hyperparathyroidism	50%	RET; 10q11.2; AD; tyrosine kinase receptor	<5	3
VHL type 2	RCC and renal cysts, central nervous system and retinal haemangioblastomas, pancreatic cysts, epididymal cystadenomas, endolymphatic sac tumours	10–20%	VHL; 3p25–26; AD; pVHL19/pVHL30	2–11	5
Neurofibromatosis type 1	Neurofibromas, café-au-lait spots	1%	NF1; 17q11.2; AD; neurofibromin	Unknown	11
PGL-4	Carotid body tumours; vagal, jugular, tympanic, abdominal, thoracic paragangliomas	$\approx 20\%$	SDHB; 1p36.13; AD; catalytic iron-sulphur protein	3–10	30–50
PGL-1	As PGL-4	$\approx 20\%$	SDHD; 11q23; maternal imprinting ; CybS membrane subunit	4–7	<3

Data from Dluhy RG, 2002; Lenders JW et al., 2005. AD, autosomal dominant; PGL, familial paraganglioma syndrome.

- **What it tests:** malignancy is low in MEN-2 and VHL but **high only with SDHB**; MEN-2 carries the highest phaeochromocytoma risk. **Trap:** SDHD is maternally imprinted, not simply dominant.

Malignant pheochromocytoma

- **Malignancy is defined only by clinical metastases**; histology and even local invasion predict poorly; no single histological criterion; **Ki-67** may be the best marker but is unvalidated.
- Rare in MEN-2 and VHL; **SDHB is the mutation most strongly associated with malignancy**.

Clinical Characteristics*Overview***THE “10% TUMOUR” RULE VERSUS CONTEMPORARY DATA**

Classic 10%	Contemporary
Extra-adrenal	Up to 25%
Familial	Up to 30% (about one-third)
Malignant	Sporadic adrenal up to 5%; most hereditary low; >one-third of extra-adrenal
Paediatric	Commonest endocrine neoplasm of childhood; up to 40% familial, >20% bilateral
Bilateral	Commoner in hereditary disease

- Sporadic cases peak in the **4th–5th decades**; familial cases present younger. **Right-sided tumours** are commoner, larger and recur more.
- **Paroxysmal hypertension is classic but seen in only 30–50%**; the rest have sustained hypertension, a minority are normotensive. **Classic triad: headache, episodic sweating, tachycardia. >20% asymptomatic.**
- In a Mayo Clinic autopsy series, **>75% of tumours were unsuspected in life.**

TABLE 146.7 CLINICAL MANIFESTATIONS OF PHAEOCHROMOCYTOMA

Frequency band	Manifestations
Highest (≈50–90%)	Headache 60–90% ; sweating 55–75%; palpitations 50–70%; sustained hypertension 50–60%
Intermediate (≈20–45%)	Pallor 40–45%; hyperglycaemia 40%; nausea, weight loss, tiredness, anxiety/panic 20–40%; paroxysmal hypertension 30%
Variable/lower	Orthostatic hypotension 10–50%; flushing 10–20%

Modified from Lenders JW et al., 2005.

- **Trap: sustained hypertension is commoner than paroxysmal hypertension**; flushing is uncommon (pallor is the rule).

Hereditary disease

- **Younger, multifocal and/or bilateral. MEN-2: almost always adrenal**; VHL ≈12% and NF-1 ≈6% extra-adrenal; **SDHB/SDHD mostly extra-adrenal and multifocal**, though solitary adrenal tumours occur.

Malignant disease

- **Metastasis much commoner in extra-adrenal tumours**; with SDHB, a substantial minority (≈28%) present with adrenal lesions.
- Malignant tumours tend to be **larger (>5 cm) and dopamine secreting** — but no preclinical diagnosis without metastasis.
- **Sites: bone, lung, liver, lymph nodes.** Most metastases appear **within 5 years**, but **>15 years** is reported.

Diagnostic Tests

- **Biochemistry first, then imaging to localise.** Consider pheochromocytoma even in **known malignancy** and **suspected solitary adrenal metastasis** — misdiagnosis can be catastrophic.

Cross-sectional imaging

- **Well circumscribed; >10 HU unenhanced (mean ≈35 HU)** — rich vascularity, low lipid; **no rapid washout** (rare <10 HU, brisk-washout tumours exist → biochemistry for every mass).

- **Iodinated contrast does not trigger hypertensive crisis** — a discredited belief.
- **MRI**: no out-of-phase signal drop-out. The “**light bulb sign**” (bright T2, best on fat suppression) is **neither sensitive nor specific**.

Functional imaging

- **¹⁸F-FDG PET, ⁶⁸Ga-DOTATATE PET/CT and MIBG** — see Imaging of Adrenal Masses.

Biochemical Evaluation

- Catecholamines and metabolites circulate largely **sulphate-conjugated**; old “total” assays could not separate free from sulphated forms and performed worse.
- **Catecholamine testing** (sensitivity and specificity ≈85%) has been replaced by metanephrines because **release is paroxysmal**; urinary catecholamines are still often obtained with urinary fractionated metanephrines.
- **Metanephrine testing — the foundation**: COMT converts catecholamines to metanephrines **continuously inside the tumour**, so plasma metanephrines are **far more sensitive than episodic catecholamine surges**. Choice of plasma versus urine remains debated.
- **VMA**: formed via MAO in both medulla and sympathetic nerves (which lack PNMT and add only normetanephrine), so the rise is diluted → **low sensitivity (<65%) but high specificity (≈99% in non-familial)**.
- **Clonidine suppression**: α2 agonist suppresses sympathetic (not tumour) norepinephrine; compare normetanephrine before and after — secondary test for **borderline** results; **risk of significant hypotension**.
- **Chromogranin A**: secretory-vesicle granin; sensitivity ≈85%; role in confirmation of mild (<4-fold) plasma metanephrine rises; **renally cleared — specificity falls when GFR <80 mL/min**.

TABLE 146.8 TEST CHARACTERISTICS FOR DIAGNOSIS OF PHAECHROMOCYTOMA (LARGE MULTICENTRE COHORT)

Test	Sensitivity hereditary	Sensitivity sporadic	Specificity hereditary	Specificity sporadic
Plasma free metanephrines	97	99	96	82
Plasma catecholamines	69	92	89	72
Urinary fractionated metanephrines	96	97	82	45
Urinary catecholamines	79	91	96	75
Urinary total metanephrines	60	88	97	89
Urinary VMA	46	77	99	86

From Lenders J et al., JAMA 2002 (values %).

- **What it tests: fractionated metanephrines win on sensitivity** (best to exclude); **VMA and total metanephrines win on specificity but miss tumours**. **Trap**: in the same cohort urinary fractionated metanephrines had poor specificity in sporadic disease, whereas the tandem mass spectrometry series cited earlier reports ≈91% — different assay methods and populations.

Screening for hereditary phaeochromocytoma

- Hereditary in more than one-third; nearly one-quarter of apparently sporadic cases carry germline mutations → **all patients with phaeochromocytoma or paraganglioma should enter shared decision-making about genetic testing** (2014 Endocrine Society guideline algorithm).

Treatment

Overview

- **Phaeochromocytoma is a surgical disease:** complete resection whenever possible; **laparoscopic adrenalectomy is standard**, open for large or difficult tumours. **Lifelong follow-up for all;** familial and malignant disease need cardiology, endocrinology and oncology input.

Preoperative management

- Intraoperative catecholamine release → hypertensive crises and arrhythmias; **historical mortality up to 50%** before routine blockade, now **<3%** (anaesthesia plus blockade).
- **First International Symposium on Pheochromocytoma (2005): preoperative blockade for all with abnormal biochemistry, even if normotensive and asymptomatic.** No level 1 evidence on the best regimen.
- Two strategies (Fig. 146.22): **option 1 — aggressive α -blockade (NIH approach); option 2 — calcium channel blockade.** A minority advocate no blockade in selected patients.
- **Cardiac evaluation** for catecholamine cardiomyopathy — some advise routine echocardiography; at least cardiology or anaesthetic consultation.

PREOPERATIVE DRUGS FOR PHAECHROMOCYTOMA (FIG. 146.22)

Class / agent	Mechanism	Dosing	Key points
Phenoxybenzamine (option 1, first line)	Irreversible non-selective α-blockade — surges cannot override it; reversal needs new receptor synthesis	Start 7–14 days preop: 10 mg twice daily, $\uparrow$ 10–20 mg every 2–3 days; $\approx$ 1 mg/kg total usually enough. Children: 0.2 mg/kg (max 10 mg) four times daily, $\uparrow$ by 0.2 mg/kg	Last dose night before surgery, omit morning dose; post-resection hypotension from lasting blockade; causes false-positive metanephrines
Doxazosin, terazosin, prazosin	Selective reversible α_1 -blockade	—	Fewer side effects; efficacy data contradictory
Atenolol 12.5–25 mg two–three times daily; metoprolol 25–50 mg three–four times daily	Selective β_1 -blockade	Only for tachycardia/arrhythmia	Never before adequate α-blockade
Metyrosine (α -methyl-p-tyrosine)	Inhibits tyrosine hydroxylase — rate-limiting step (tyrosine $\rightarrow$ L-DOPA)	250 mg twice–three times daily, $\uparrow$ 250–500 mg every 2–3 days; 1.5–2 g/day usually enough; 500 mg night before surgery; full effect $\approx$ 3 days	Incomplete block $\rightarrow$ combine with α -blockade; institutional preference. CNS effects: sedation, depression, galactorrhoea, extrapyramidal symptoms $\rightarrow$ stop metyrosine ; some reserve for refractory/metastatic disease
Calcium channel blockers (option 2, or add-on): amlodipine 10–20 mg, nicardipine 60–90 mg, nifedipine 30–90 mg, verapamil 180–540 mg daily	Smooth muscle relaxation	Sole therapy (option 2) or added if blood pressure still poorly controlled	Avoids reflex tachycardia and postoperative hypotension; NIH reserves sole use for mild symptoms

- **Why never β before α :** unopposed β -blockade removes β_2 arteriolar dilatation, so epinephrine acts unopposed on α_1 receptors $\rightarrow$ hypertensive crisis. Hence **selective β_1 agents**.
- **Targets (outpatient, 7–14 days):** seated **120–130/80 mmHg**; mild postural hypotension acceptable — systolic **>80 mmHg** (text) or **>90 mmHg** (Fig. 146.22); a sign of adequate blockade.

- **Normotensive asymptomatic patients:** in one cohort, no blockade gave equivalent perioperative outcomes, and doxazosin-treated patients needed **more** intraoperative vasoactive drugs — needs prospective validation.
- **Trap:** extrapyramidal side effects belong to metyrosine, not phenoxybenzamine.

Intravascular volume management

- **Perhaps the most important preoperative step.** Liberal salt and fluid once blockade starts → admit the day before for **IV fluids (1–2 L bolus the night before)**.

Postoperative management

- **Hypotension** (lasting phenoxybenzamine effect) and **rebound hypoglycaemia:** catecholamine excess suppressed insulin via α_2 receptors; withdrawal → rebound hyperinsulinaemia. Some advise **overnight intensive care**.

Follow-up

- **Repeat biochemistry after surgery;** if metanephrines stay high, **MIBG** may unmask metastases once the primary is removed.
- **Annual biochemistry is mandatory; lifelong screening advised — 10-year recurrence up to 16%,** even >15 years later. Suggested: **testing at 6 months, then annually;** postoperative imaging to document resection, further imaging guided by biochemistry.

Hereditary disease

- MEN-2 and VHL: **low malignancy, high bilaterality → partial cortical-sparing adrenalectomy** to avoid lifelong steroid replacement. Requires expertise and planning; replacement may still be needed, **addisonian crisis can still occur;** recurrence poorly defined but significant.

Malignant disease

- **Largely palliative; metastasectomy of resectable disease is standard** but unproven for survival versus α/β -blockade plus α -methyl-p-tyrosine. Ablation and embolisation for local control.
- **^{131}I -MIBG therapy** (confirm uptake first): symptomatic response in up to two-thirds; catecholamine fall in >40%; tumour shrinkage $\approx$ 30%; complete response <5%. High-dose therapy produced durable complete responses in a quarter, with notable toxicity. **FDA approval 2018 (Azedra)** for unresectable pheochromocytoma/paraganglioma, from a single-arm open-label study.
- **Cyclophosphamide, vincristine, dacarbazine (CVD):** best-studied chemotherapy; >50% radiological and $\approx$ 75% biochemical response, but short-lived ($\approx$ 2 years); used after MIBG failure or MIBG-negative tumours. CVD + MIBG undefined.

Prognosis

- **Excellent after resection; surgical mortality <3%;** hypertension usually but not always resolves.
- **Recurrence up to 16%,** half malignant; risk factors: **extra-adrenal (hazard ratio 11.2), hereditary (3.4), right-sided (3.1), bilateral (1.4), larger (1.2 per cm).**
- Metastatic course varies — some survive >20 years; **bone metastases carry the best prognosis; 5-year survival $\approx$ 50%.**
- **Summary:** consider pheochromocytoma in every adrenal mass; refer to a tertiary centre if unfamiliar with its care.

KEY POINTS: PHAEOCHROMOCYTOMA

- **Nearly 25% of apparently sporadic cases are hereditary** on genetic testing.
- **Plasma or urinary metanephrines** underpin modern diagnosis.

- **Malignancy is defined only by metastases**; recurrence even 15 years after resection → long-term follow-up.
- **Master perioperative management** before operating.

Concept — Every step of pheochromocytoma care follows from one fact: the tumour stores and releases catecholamines unpredictably but metabolises them continuously. Continuous metabolism makes metanephrines the diagnosis; unpredictable release makes α -before- β blockade and volume repletion the preparation; and delayed metastasis makes follow-up lifelong.

NONFUNCTIONAL BENIGN ADRENAL LESIONS

Myelolipoma

Epidemiology

- **Rare, benign, metabolically silent**; all ages; <0.1% at autopsy. About 15% extra-adrenal, up to half of these **presacral**.
- **Equal sex ratio; no side preference**; bilateral cases reported; may exceed 10 cm; usually asymptomatic, occasionally abdominal or flank pain.

Pathophysiology

- **Clonal stem-cell proliferation**: mature fat plus **trilineage haematopoietic tissue** (megakaryocytes may abound). Cellularity does not fall with age; **no contribution to circulating blood** — capillary venous sinuses are absent.
- Induced in rats by necrotic tissue and ACTH → the **adrenal-stress-response hypothesis**; cause unknown.

Diagnosis

- **Large macroscopic fat component** on cross-sectional imaging; interdigitating enhancing myeloid tissue; calcification in over a quarter; haemorrhage common. Fig. 146.24: **macroscopic fat in a 1.5-cm lesion is pathognomonic**.

Additional Workup

- **Hormonal testing recommended for all**, although some guidelines advise against it: over two dozen myelolipomas coexist with hormone excess, **usually hypercortisolaemia**, generally from a **coexisting functional adenoma** rather than the myelolipoma itself.

Management

- **Surgical excision is definitive when needed**; embolisation only for acute haemorrhage. **Most do not need resection** (Fig. 146.23: a rare 15-cm symptomatic lesion).
- Indications: **symptoms**, or concern about **spontaneous haemorrhage** (fewer than 50 cases; smallest bleeding lesion 6.3 cm; bleeders average 10–14 cm). No deaths reported; emergency surgery or transfusion rare → **value of prophylactic excision unclear — individualise**.

PROPOSED SIZE CUT-OFFS FOR ELECTIVE MYELOLIPOMA EXCISION

Threshold	6 cm	7 cm	8 cm	10 cm
Basis	Different series; none validated			

- **No follow-up guidelines**; in small series only half of unresected lesions grew.

Adrenal Cysts

Overview and Epidemiology

- 0.064–0.18% at autopsy; **1–22% of incidentalomas**. Millimetres to >20 cm; uni- or multilocular; **bilateral in 8–10%; female predominance, peak 3rd–6th decades**; from prenatal life onwards.
- Associations: hypertension, polycystic kidney disease, **Klippel–Trénaunay–Weber** and **Beckwith–Wiedemann** syndromes.

Pathophysiology

HISTOLOGICAL TYPES OF ADRENAL CYST

Type	Share	Features
Endothelial	45%	No proliferating endothelium; lymphangiomatous and angiomatous subtypes
Pseudocyst	39%	No cellular lining; after intra-adrenal haemorrhage or infarction ; possibly related to endothelial cysts; the type associated with malignancy
Epithelial	9%	True epithelium: glandular, embryonal, cystic adenoma
Parasitic	7%	Disseminated <i>Echinococcus</i> ; rarely the sole site

Clinical Characteristics

- **Cystic neoplasms mimic benign cysts**, especially haemorrhagic or degenerate tumours resembling pseudocysts (ACC, adenoma, pheochromocytoma). **About 7% of reported cysts were associated with malignancy — all pseudocysts**; likely overestimated because series are surgical.
- Cystic neoplasms are **larger (>7 cm) with thicker walls**. Large cysts may be hard to assign to adrenal versus kidney, pancreas, spleen or liver.

Diagnostic Tests

- **Endocrine evaluation for all.**
- Imaging: **well-defined, sharply margined, fluid attenuation, no enhancement** (Fig. 146.25: stippled wall calcification). Attenuation **>30 HU** suggests haemorrhage, debris or calcification. **Peripheral, rim-like calcification in 15–70%** (occasionally nodular). MRI: T1 hypointense, T2 hyperintense.
- Imaging cannot exclude malignancy → aspiration or excision. **Aspirate only after excluding pheochromocytoma**; send fluid for **Gram stain, cytology, triglycerides, cortisol and amylase (left-sided)**; cystogram can assess the wall. Aspiration may be therapeutic.

Treatment

- Driven by **function, malignancy risk and symptoms**. Natural history: in small series 60% enlarged and 25% shrank.
- **Observe** small, asymptomatic, thin-walled, homogeneous near-water cysts; ESE–ENSAT: **no follow-up for silent benign-appearing lesions <4 cm**.
- **Excise** cysts that are **heterogeneous, >5 cm, thick-walled or symptomatic**. Simple decortication is not advocated (malignancy risk).

Prognosis

- Histology dependent: benign cysts are followed for reaccumulation and retreated if symptoms recur; malignant ones per their histology.

Summary

- **Symptomatic cysts → surgery; small, non-functional, asymptomatic, benign-appearing → conservative follow-up.**

KEY POINT: ADRENAL CYSTS

- Because of possible associated malignancy, observe adrenal cysts with caution.

Concept — The cyst that carries risk is the pseudocyst, because a haemorrhagic or necrotic tumour can masquerade as one. Size, wall thickness and heterogeneity are therefore surrogates for “is there a tumour behind the fluid?”.

Ganglioneuroma

Overview and Epidemiology

- Extremely rare benign neuroectodermal tumour of ganglion and Schwann cells; 0.3–2% of incidentalomas.
- Sites: **mediastinum ≈39%, retroperitoneum ≈31%, adrenal ≈21%**, occasionally pelvis and neck. **Young:** half aged 10–29; only 20% over 40.

Pathophysiology/Clinical Characteristics

- Rare malignant transformation; peripheral nerve sheath tumours and composite ganglioneuroma–phaeochromocytoma reported; nodal ganglioneuroma cells may represent **matured neuroblastoma**.
- **Can grow very large and encase vessels without narrowing the lumen.** May secrete **vasoactive intestinal polypeptide** (diarrhoea); most are asymptomatic.

Diagnostic Tests

- **No definitive preoperative diagnosis;** imaging clues: **well circumscribed; attenuation ≤ muscle** on enhanced and unenhanced CT; **calcification in 20–69%** (classically fine and punctate, often coarse; limited value because ACC also calcifies).
- MRI: **T1 homogeneously low; T2 heterogeneous, intermediate to high.** PET standardised uptake value usually <3.0.

Treatment and Prognosis

- Diagnosis almost always at resection, so natural history is undefined; beware **large size and encasement of retroperitoneal vessels**.

Summary

- Rare, largely pathological diagnosis, **nearly always benign course**.

KEY POINT: GANGLIONEUROMA

- Rare benign tumours of the young that can grow to encase critical structures.

Adrenal Oncocytoma

Overview and Epidemiology

- Familiar from the kidney; also salivary and endocrine glands. **Adrenal oncocytomas are exceedingly rare** (about 50 reported).

Pathophysiology and Clinical Characteristics

- **Large, mitochondria-rich, granular eosinophilic cells;** pathogenesis unclear.
- **Female:male ≈2.5:1; left:right ≈3.5:1;** up to 20 cm; cases in pregnancy.
- Mostly inactive; **≈10% secrete** sex steroids, cytokines and/or cortisol.

- Classically benign, but **>30% of reported lesions malignant** and a further 23% indeterminate, including IVC invasion.

Diagnostic Tests and Treatment

- Diagnosed at resection → **manage like any adrenal mass. No regular central stellate scar** (unlike renal oncocytoma). Metastasectomy and radiotherapy for local recurrence may help malignant cases.

Prognosis

- Uncertain but **leans benign**; even widely metastatic disease may progress slowly.

Summary

- Extremely rare; despite a benign reputation, **a proportion are malignant**.

KEY POINT: ONCOCYTOMA

- **Unlike renal oncocytomas, a large proportion of adrenal oncocytic lesions are malignant.**

Concept — *The benign-sounding adrenal lesions are benign only once imaging or pathology has proved it. Macroscopic fat alone is diagnostic — hence myelolipoma alone escapes the 4-cm rule — while cysts, ganglioneuromas and oncocytomas are managed as adrenal masses until resected.*

MALIGNANT ADRENAL NEOPLASMS

Adrenocortical Carcinoma

Overview and Epidemiology

- **Rare: 0.7–2 per million. Bimodal** — children in the first decade, adults in the **4th–5th decades** (most patients are adults); **slight female predominance (1.5–2.1:1)**.
- **Mostly sporadic and unilateral; 2–6% bilateral**, possibly hereditary.
- Associated syndromes: **Li-Fraumeni, Beckwith-Wiedemann**, Lynch, Carney complex, MEN-1, NF-1, familial adenomatous polyposis, McCune-Albright.
- **Surgery is the cornerstone and the best chance of cure**; advanced disease needs multimodal therapy.
- Despite more incidental imaging, there has been **no stage or size migration** and **no survival improvement** over two decades; administrative data show only a **weak relationship between size and metastasis or survival**.

Pathophysiology

MOLECULAR DRIVERS: ACC VERSUS ADENOMA

Alteration	Syndrome model	Sporadic ACC	Paediatric ACC	Adenoma
Loss of TP53	Li-Fraumeni	20–33%	50–80%	0–6%
↑ Insulin-like growth factor (IGF)	Beckwith-Wiedemann	Up to 90%	—	8.5%

- Both are regarded as **late events** in sporadic tumorigenesis.
- **The Cancer Genome Atlas**: commonest alterations **TP53, ZNRF3, CTNNB1, CDKN2A, TERT, PRKAR1A**. **CpG island hypermethylation phenotype (CIMP)** defines three prognostic subtypes — **5-year survival ≈80% (CIMP-low), 43% (intermediate), 0% (high)**. Genomic diversity suggests **multi-pathway targeting** is needed.

Clinical Characteristics

- **Most still present advanced and symptomatic:** hormone excess 40–60%, abdominal pain/fullness ≈30%, remainder incidental.
- **Functional in 50–79% of adults and 90% of children;** functional status may not cause symptoms (e.g., steroid precursors) and may change as the tumour grows; several hormones are often co-secreted.
- **Cortisol is commonest** (50–80% of functional tumours) — **ACTH-independent Cushing syndrome**, often with more marked virilisation than functional adenomas; ACC causes ≈8% of Cushing syndrome; **cortisol secretion independently predicts poor outcome.**
- **Androgens second** (40–60% of functional tumours): in women, **virilisation** (male-pattern baldness, hirsutism, oligomenorrhoea) related to 17-ketosteroids; virilisation **without** raised 17-ketosteroids → consider testosterone from adrenal or ovary.
- **Oestrogen → feminisation in men** (testicular atrophy, gynaecomastia), partly peripheral conversion of androstenedione — **highly suggestive of malignancy.**
- **Cortisol plus androgen** in up to half of functional ACCs. **Aldosterone is rare** — hypertension and hypokalaemia more often come from other steroids.
- **Non-functional ACC:** fullness, back pain, nausea, vomiting, constitutional symptoms.

TABLE 146.9 FUNCTIONAL BREAKDOWN OF ADRENOCORTICAL CARCINOMAS

Status	Proportion (table)	Body text
Non-functional	21–50%	—
Functional	50–79%	Hormone symptoms 40–60% of patients
— Cushing syndrome	33–53%	Cortisol in 50–80% of <i>functional</i> tumours
— Cushing plus virilisation	20–24%	Up to 50% of functional tumours
— Virilisation alone	10–20%	Androgens in 40–60% of functional tumours
— Feminisation	6–10%	1–3% of patients
— Hyperaldosteronism	2.5–5%	Rare

- **Why values differ:** the table uses **all ACCs** as denominator and clinical syndromes; the text quotes proportions of **functional** tumours and biochemical secretion, from different series. **Pattern:** cortisol dominates, androgens next, oestrogen and aldosterone rare. **Trap:** feminisation in a man with an adrenal mass means carcinoma until proved otherwise.

Diagnostic Tests

- Suspicion from symptoms or imaging → **full extent-of-disease evaluation**, history and examination for hormone excess.
- **Functional evaluation is essential** — for diagnosis, planning **postoperative cortisol replacement**, and **tumour markers for surveillance** (Table 146.10).

TABLE 146.10 FUNCTIONAL EVALUATION OF ADRENAL TUMOURS SUSPECTED TO BE ACC

Hypersecretion	Tests
Glucocorticoid	LD-DST (or LNSC or 24-h urine cortisol); ACTH
Sex steroids and precursors	DHEA-S, testosterone ; optional 17-OH-progesterone, androstenedione, 17-OH-pregnenolone, 11-deoxycorticosterone, progesterone, oestradiol; 17β-oestradiol only in men and postmenopausal women
Catecholamine	Serum or urinary metanephrines
Mineralocorticoid	ARR — only with hypertension or hypokalaemia

Modified from Fassnacht M et al., 2004.

- **Size:** average **10–12 cm**; **>90% exceed 5 cm**; carcinoma in 4–5% of incidentalomas <4 cm, 10% >4 cm, 25% >6 cm → **consider resection above 4–6 cm**.
- **Ominous:** lymphadenopathy, local extension into IVC or adjacent organs. **Commonest metastatic sites: lung and liver** → **CT chest, abdomen, pelvis**; bone and brain imaging only if symptomatic.
- **CT:** irregular borders and enhancement, calcification, necrosis with cystic change; **unenhanced mean ≈39 HU vs ≈8 HU in adenoma**; classically **no brisk washout** (exceptions reported).
- **MRI:** **T1 isointense to liver/spleen, T2 intermediate–high, marked gadolinium uptake**; **essential for venous tumour thrombus**.
- **¹⁸F-FDG PET:** ACC is avid (high sensitivity and specificity for malignancy) but **cannot distinguish ACC from other malignancies or metastases**.
- **No percutaneous biopsy before resection** — needle-tract seeding; biochemistry and imaging justify surgery. **Biopsy only for unresectable, locally advanced or metastatic disease** before systemic therapy.

Figure legend — Fig. 146.26

- A 25-cm stage II right ACC **displaced the kidney anteromedially** (renal masses of similar size usually do not) and looked deceptively **homogeneous and well circumscribed**.

Pathological Evaluation

- Two questions: **origin** (steroidogenic factor 1 (SF1) staining = adrenocortical) and **malignancy** — obvious with metastasis or invasion, difficult when organ confined (Fig. 146.27: ACC — high-grade atypia with necrosis; adenoma — orderly trabeculae, minimal atypia; pheochromocytoma — alveolar/trabecular, granular amphophilic cytoplasm; metastasis — pleomorphic hyperchromatic cells).
- **Weiss criteria (1984):** nine features of structure, cytology and invasion; **≥3 criteria = malignant** (sensitivity 100%, specificity 96%). **Mitotic rate** grades ACC (low vs high) and correlates best with outcome. **Use caution in paediatric and oncocytic tumours** — alternative criteria recommended.

BOX 146.6 WEISS PATHOLOGICAL CRITERIA FOR DIFFERENTIATING BENIGN AND MALIGNANT ADRENAL TUMOURS

Domain	Criteria
Cytology	High nuclear grade (Fuhrman 3–4); >5 mitoses per 50 high-power fields ; atypical mitoses
Structure	Low percentage of clear cells ; diffuse architecture ; necrosis
Invasion	Venous, sinusoidal, capsular

From Weiss LM, 1984.

- **Mnemonic of the three domains:** cells, architecture, invasion — three of nine positive = carcinoma.

Staging

AJCC VERSUS ENSAT STAGING OF ACC

Stage	American Joint Committee on Cancer (AJCC)	ENSAT
I	T1 (≤5 cm) N0 M0 — confined	Same
II	T2 (>5 cm) N0 M0 — confined	Same
III	Extension into periadrenal fat (T3) or regional nodes (T1–2 N1)	N1, T3, T4 or venous tumour thrombus
IV	Invasion of adjacent organs (T4), T3N1, or distant metastasis	Distant metastasis only

TABLE 146.11 STAGING OF ACC WITH STAGE AT DIAGNOSIS AND 5-YEAR SURVIVAL (2004 UICC/WHO)

Stage	TNM	At diagnosis	5-year survival
I	T1N0M0	3–4%	33–66%
II	T2N0M0	29–46%	20–58%

Stage	TNM	At diagnosis	5-year survival
III	T1–2N1M0; T3N0M0	11–19%	18–24%
IV	T1–4N0–1M1; T3N1M0; T4N0–1M0	39–49%	<5%

T1 ≤5 cm; T2 >5 cm; T3 infiltration of surrounding fat; T4 invasion of adjacent organs. UICC, Union Internationale Contre le Cancer; WHO, World Health Organization.

- **Pattern:** stage I is rare at diagnosis and **the largest group is stage IV**. **Trap:** ENSAT moves T4 and N1 disease without metastasis into stage III, reserving stage IV for distant spread.

Management

- Most present advanced; localised disease still has **high local and distant recurrence (60–80%)** → **multidisciplinary, multimodal care**. Only one randomised trial exists in metastatic disease; the rest is retrospective.

Surgery

- **Complete excision is essential** — best chance of cure, control of hormonal and local symptoms.
- **Open surgery preferred for tumours >6 cm or with local invasion** (higher peritoneal seeding with minimally invasive approaches); smaller organ-confined tumours may be approached laparoscopically with caution (spillage risk; outcomes may favour open).
- **En bloc resection** of involved organs for negative margins; **regional lymphadenectomy** for control and staging; **IVC thrombus** may need bypass, replacement or interruption.
- **Metastatic disease: cytoreduction if >90% of tumour burden can be removed** — may palliate and facilitate other therapy, survival benefit uncertain. **If complete debulking is impossible, do not operate** — it delays systemic therapy.
- **Resect recurrences** (improved survival in retrospective series), best in **low-grade tumours with long disease-free intervals**; radiofrequency ablation or angioembolisation if unresectable.
- **Surveillance:** CT chest, abdomen, pelvis **every 3 months for 2–3 years → every 6 months to 5 years**; monitor initially raised hormones; PET for indeterminate findings; beyond 5 years individualised (<3% recur after 5 years).

Radiation

- **Treatment of choice for bone and brain metastases**; limited role for the primary.
- **Adjuvant radiotherapy lowers 5-year local recurrence (≈86% without vs ≈11% with) but not disease-free or overall survival**; best for patients **without metastasis with positive or indeterminate margins**.

Medical therapy

- **Mitotane** — oral dichlorodiphenyltrichloroethane (DDT) derivative, the most used agent: **adrenolytic** (direct cortical cytotoxicity, free-radical damage, steroidogenic enzyme inhibition) → **steroid replacement needed**. Toxicity: gastrointestinal upset, lethargy, depression, feminisation in men, rash, raised liver enzymes and binding globulins, hypothyroidism, dyslipidaemia, thrombocytopenia → **titrate to tolerance and serum levels**.
- Single-agent response 14–36%; streptozotocin combination ≈36%; etoposide–doxorubicin–cisplatin combination ≈49%.
- **Symptom control** while mitotane takes effect: **ketoconazole**, **metirapone**, **aminoglutethimide**, **etomidate** (adrenostatic) and **mifepristone** (glucocorticoid receptor antagonist).

KEY STUDIES OF SYSTEMIC THERAPY IN ACC

Study	Setting and comparison	Result	Verdict
Large retrospective series (adjuvant)	Adjuvant mitotane vs none after resection	Improved recurrence-free and overall survival after adjustment; acceptable toxicity	Supportive but not definitive; patient selection, dose and duration unresolved
ADJUVO (randomised)	R0-resected stage I–III ACC: adjuvant mitotane vs observation (1:1)	Recurrences 8 vs 11 — not significant	No routine adjuvant mitotane , sparing toxicity
FIRM-ACT (randomised)	Locally advanced/metastatic: mitotane + etoposide, doxorubicin, cisplatin (M-EDP) vs mitotane + streptozotocin	Overall survival similar; M-EDP better response rate and progression-free survival	M-EDP is first-line systemic therapy

- **Trap:** FIRM-ACT improved response and progression-free survival, **not overall survival**; ADJUVO tested low-risk resected disease, not all patients.

Prognosis

- **Overall 5-year survival 20–47%** — advanced presentation, frequent recurrence, weak systemic therapy.
- Adverse factors: **stage, size >12 cm**, age, **positive margins**, high mitotic rate, atypical mitoses, **cortisol production**, necrosis, **high Ki-67**. The effect of functional status overall is debated.

Pediatric Adrenocortical Carcinoma

PAEDIATRIC VERSUS ADULT ACC

	Paediatric	Adult
Function	>90% functional; virilisation commonest (55–70%)	50–79% functional; cortisol commonest
Sex ratio	Equal <12 years; 6:1 female aged 13–20	Slight female predominance
Genetics	Consider Li-Fraumeni (high TP53 rate)	Mostly sporadic
Staging	I–II: completely excised, negative margins, split at 200 g ; III: residual or inoperable; IV: haematogenous metastasis	Size (5 cm), invasion, nodes, metastasis
Stage and outcome	Lower stage, better survival; 5-year ≈54%	Mostly advanced; 20–47%
Adverse factors	Weight >400 g, stage, size >10.5 cm, necrosis, frequent/atypical mitoses, age >5 years	Size >12 cm, margins, mitoses, cortisol, Ki-67

Summary

- Consider ACC in every adrenal tumour, **particularly >6 cm**; **assess function even if asymptomatic**; resect primary and metastases whenever possible; consider adjuvant therapy or trials; **mitotane alone or with cytotoxics first line for metastatic disease**; prognosis remains poor.

KEY POINTS: ADRENOCORTICAL CARCINOMA

- **Increasing size raises malignancy risk.**
- **Most ACCs are functional at presentation.**
- **Multimodal therapy** — surgery, radiotherapy, systemic therapy — is often needed.

Concept — ACC is a disease of recurrence rather than of the primary: even complete resection is followed by relapse in most patients, so outcome depends on an intact, unspilled first operation. That is why open en bloc surgery, no preoperative biopsy and hormonal markers for surveillance dominate its management.

Neuroblastoma

- **Neural-crest malignancy** (adrenal medulla, sympathetic ganglia); **commonest extracranial solid tumour of childhood** — covered in Chapter 63.

Adrenal Sarcoma

- **Extremely rare primary sarcomas:** epithelioid angiosarcoma, leiomyosarcoma, liposarcoma, Kaposi sarcoma, alveolar and synovial sarcoma.
- **Cannot be distinguished preoperatively** from other adrenal malignancies (benign adenoma is excluded by the usual work-up); definitive diagnosis can be hard even after resection.
- **Wide local excision with negative margins;** mixed carcinoma–sarcoma features → treat as **sarcomatoid carcinoma**.

Metastases

Overview and Epidemiology

- **Common site:** at autopsy, **>25% of patients with melanoma or lung carcinoma** have adrenal metastases; many other primaries (RCC, breast, medullary thyroid, contralateral ACC, gastrointestinal, prostate, cervix, pancreas, urothelial, seminoma, others) also spread there.
- **With a history of malignancy, >50% of new adrenal lesions are metastases.**
- **RCC** reaches the adrenal directly or haematogenously, yet **ipsilateral adrenalectomy at radical or partial nephrectomy only if the gland is radiologically abnormal or thrombus reaches the adrenal vein.**
- **Resection of an isolated metastasis may improve survival in melanoma and non-small cell lung cancer (NSCLC);** a randomised trial now supports metastasectomy in lung cancer.

Pathophysiology and Clinical Characteristics

- **Adrenal insufficiency from metastases is exceedingly rare** (though Addison described it); **bulky (>4 cm) bilateral disease** seems necessary. The oft-quoted “20% of metastatic RCC have adrenal insufficiency” rests on a single small abstract. Advanced cancer often mimics addisonian symptoms without biochemical insufficiency.

Diagnostic Tests

- Metastases look like adenomas (well circumscribed, homogeneous, rarely necrotic) but **lack lipid: >10 HU and poor washout** (Fig. 146.28: 38-HU left adrenal RCC metastasis years after right nephrectomy).
- **Exceptions:** up to 7% of metastases are <10 HU; **RCC and hepatocellular carcinoma metastases can wash out like adenomas** → **growth kinetics become decisive;** MRI signal intensity may help separate RCC metastasis from adenoma. **Biopsy helps** with a known primary.
- **Unlike renal metastases** (usually with widespread disease), **adrenal metastases may be early and solitary** — making resection feasible.
- **Metabolic work-up still required** — functional lesion rates are similar with or without prior malignancy, and **pigmented pheochromocytoma has been mislabelled as metastatic melanoma.**

Treatment and Prognosis

- **RCC:** resection of ipsilateral or contralateral adrenal metastasis as part of **cytoreduction in selected patients;** benefit unproven prospectively; role may change with targeted and immunotherapy.
- **Non-urological primaries:** multidisciplinary decisions with the medical oncologist; best data for **NSCLC and melanoma.**

- **NSCLC:** adrenal metastases usually accompany systemic disease; isolated deposits may reflect lymphatic spread. **Resection of a solitary metastasis → 5-year survival >25%;** synchronous and metachronous metastases fare equally (meta-analysis); **level 1 evidence supports metastasectomy.**
- **Melanoma:** long-term survival and rare regression after adrenalectomy — retrospective, highly selected data.
- **Summary:** suspect metastasis in any incidentaloma with a cancer history; percutaneous biopsy is safe (success depends on size, location, technique); **adrenalectomy for solitary metastasis** suits selected patients.

KEY POINTS: ADRENAL METASTASES

- **Common, but bilateral bulky (>4 cm) disease is needed for biochemical adrenal insufficiency.**
- **>50% of new adrenal lesions in patients with prior malignancy are metastases** — metabolic work-up still recommended.

Concept — *Prior malignancy flips the prior probability — most new adrenal lesions become metastases — but does not remove the need to exclude pheochromocytoma. Because adrenal deposits can be early and solitary, this is the one setting in which biopsy and resection of a metastasis are both justified.*

NON-NEOPLASTIC ADRENAL DISORDERS

Cushing Syndrome/Cortisol Excess

Pathophysiology

- **Cushing syndrome = symptomatic hypercortisolaemia.** Three groups: **exogenous** (iatrogenic — commonest), **ACTH-dependent** (extra-adrenal; up to 85% of endogenous cases), **ACTH-independent** (adrenal; relatively rare).

FIG. 146.29 CAUSES OF ENDOGENOUS CORTISOL EXCESS

Group	Cause	Share of Cushing syndrome
ACTH dependent (80–85% of endogenous)	Pituitary (Cushing disease)	≈70%
	Ectopic ACTH	≈10–15%
	Ectopic CRH	<1%
ACTH independent	Primary adrenal (adenoma, ACC, AIMAH, PPNAD)	≈10%

Clinical Characteristics

- Classic features — **central obesity, moon facies, buffalo hump, proximal weakness, easy bruising, striae** — are individually **non-specific**; the **combination** should prompt screening. Systemic features overlap the **metabolic syndrome** (Fig. 146.30: depression/psychosis, glaucoma, immunosuppression, peptic ulcer, salt and water retention, hypertension, visceral obesity, insulin resistance, protein catabolism, skin thinning, muscle atrophy, osteoporosis, urolithiasis, hypogonadotropic hypogonadism).
- **Urological relevance: hypogonadotropic hypogonadism in men** — low threshold to test men with low libido or erectile dysfunction, low testosterone and low gonadotropins; **urolithiasis in up to 50%** — test stone formers with cushingoid features. Treatment **reduces but does not normalise** stone risk.

TABLE 146.12 SIGNS AND SYMPTOMS OF CUSHING SYNDROME

Prevalence	Features
90–100%	Central obesity , moon face, facial plethora, decreased libido
70–90%	Purple striae, menstrual disturbance, hirsutism, erectile dysfunction, hypertension
50–70%	Muscle weakness, buffalo hump, bruising, glucose intolerance/diabetes, osteopenia/osteoporosis, emotional lability/depression
20–50%	Headache, backache, limb oedema, recurrent infection, hypokalaemic alkalosis, nephrolithiasis
0–20%	Acne, alopecia

Modified from Pivonello R et al., 2008. Band assignment follows the source's printed grouping.

- **Trap:** central obesity is nearly universal but least specific; the more **discriminating signs** (proximal weakness, bruising) sit lower, at 50–70%.

Diagnosis

- **Establish cortisol excess, then its cause** — see Testing for Hypercortisolism.

Exogenous Cushing Syndrome

- **Withdraw glucocorticoids gradually** to let the HPA axis recover (weeks to months). Beware **steroid withdrawal syndrome** — intolerance of dose reduction despite normal HPA testing.

Adrenocorticotrophic Hormone–Dependent Disease

- **80–85% of endogenous Cushing syndrome**; ≈80% of it pituitary (Cushing disease), the rest ectopic ACTH, and very rarely ectopic CRH.

Cushing disease

- **Harvey Cushing (1932)**; ≈70% of endogenous Cushing syndrome; usually a **microadenoma** (macroadenomas ≥1 cm ≈5%); 15% of functional pituitary adenomas secrete ACTH; up to two-thirds of patients are women.
- **Standard: trans-sphenoidal resection** — cure only **60–80%**, relapse up to 25%; **macroadenomas <15% cured**. Postoperative severe addisonian state is common (careful replacement for a year); hypopituitarism ranges widely and rises with reoperation; **salvage radiotherapy** is slow and risks hypopituitarism.
- **Bilateral adrenalectomy** (the urologist's role): usually **after at least one failed pituitary treatment**, or when **hypercortisolism is life threatening**. Advantages: very high, rapid success with no hypopituitarism. Costs: **lifelong glucocorticoid and mineralocorticoid replacement**, rare residual functioning adrenal tissue, and **Nelson (Nelson–Salassa) syndrome**.
- **Nelson syndrome**: progressive growth of the pituitary adenoma after bilateral adrenalectomy — **chiasmal compression, oculomotor deficits**, rarely raised intracranial pressure; **8–29%**; prophylactic radiotherapy inconsistent and controversial.

Ectopic ACTH syndrome

- **Nearly always malignant tumours**; ≈10% of Cushing syndrome; first described by Liddle and colleagues in the early 1960s. Hypercortisolism may precede the tumour diagnosis by years; underdiagnosed in end-stage cancer.
- **Higher ACTH → larger, more hyperplastic adrenals than in Cushing disease** (Fig. 146.32: bilateral enlargement with metastatic pancreatic adenocarcinoma).
- **Best treatment: resect the primary** — possible in only ≈10%. **Unresectable or occult primary (up to 35%) → bilateral adrenalectomy** with lifelong replacement is excellent.

TABLE 146.13 SOURCES OF ECTOPIC ACTH (RANGE ACROSS FIVE SERIES)

Source	Range (%)	Comment
Bronchial carcinoid	7–49	Commonest in most series
Small cell lung cancer	10–33	—
No tumour found	12–32	Occult source
Islet cell tumour	4–16	—
Thymic carcinoid; malignant thymoma	5–16; 13	—
Other bronchial carcinoma (adeno, squamous, undifferentiated)	3–10 each	One older series
Medullary thyroid cancer	8	—
Disseminated neuroendocrine or gastrointestinal carcinoid	1–7	—
Pheochromocytoma	3–11	Notable adrenal source
Pancreatic adenocarcinoma; miscellaneous	2; 6–17	Includes gallbladder carcinoid, colon cancer

Modified from Aniszewski JP et al., 2001 (series of 106 patients, compared with four earlier series).

- **What it tests:** ectopic ACTH is mainly **thoracic neuroendocrine** (bronchial carcinoid, small cell cancer, thymic tumours), with a substantial occult fraction — and **phaeochromocytoma can cause Cushing syndrome**.

Ectopic CRH

- **<1% of Cushing syndrome; bronchial carcinoma** commonest; concurrent ACTH secretion not unusual.

Adrenocorticotrophic Hormone–Independent Hypercortisolism

- Minority of endogenous cases: **cortisol-secreting adenoma, ACC, AIMAH, PPNAD**.

AIMAH VERSUS PPNAD

	ACTH-independent macronodular adrenal hyperplasia	Primary pigmented nodular adrenocortical disease
Share of Cushing syndrome	<1%	<1%
Glands	Massively enlarged: multiple nodules up to 4 cm; >60 g each (reports >200 g)	Normal size; black or brown cortical nodules in atrophic cortex; medulla spared
Association	McCune–Albright syndrome (polyostotic fibrous dysplasia, skin and endocrine features)	Carney complex in about half (autosomal dominant: spotty skin/mucosal pigmentation, tumours including Sertoli cell tumours); rest sporadic
Diagnostic clue	Must be distinguished from ACTH-driven hyperplasia (similar imaging)	Paradoxical delayed rise in 24-h urinary cortisol after dexamethasone
Treatment	Bilateral adrenalectomy with lifelong replacement; unilateral resection of the larger gland has succeeded in AIMAH	

Medical Treatment of Hypercortisolism

- For **bridging to surgery or when surgery is impossible**; specialist endocrinology territory.

DRUGS FOR CORTISOL EXCESS (AS NAMED IN THE CHAPTER)

Agent	Class stated	Setting in chapter
Metypapone, aminoglutethimide, trilostane, ketoconazole, etomidate	Steroid-synthesis enzyme blockers (adrenostatic)	Cushing syndrome; ACC symptom control
Mitotane	Adrenolytic	ACC; hypercortisolism

Agent	Class stated	Setting in chapter
Mifepristone	Glucocorticoid receptor antagonist	ACC symptom control

Prognosis

- **Life expectancy is reduced**; normalising cortisol lowers but does not abolish risk → **aggressive cardiovascular risk management** even after biochemical cure.

Concept — Cushing syndrome is classified by where the drive lies — outside the patient, above the adrenal, or within it — and the ACTH level locates it. The urologist's operation is either the cure (ACTH-independent disease) or the salvage (bilateral adrenalectomy when the ACTH source cannot be controlled), and the salvage exchanges hypercortisolism for lifelong replacement and Nelson syndrome.

Adrenal Insufficiency

Overview and Epidemiology

- **Addison disease**; prevalence **39–60 per million**. Relevant because it follows **resection or pharmacological inhibition** of adrenal tissue; **unanticipated addisonian crisis after simultaneous or staged bilateral adrenalectomy can kill**.

Pathophysiology

PRIMARY VERSUS SECONDARY ADRENAL INSUFFICIENCY

	Primary (adrenal)	Secondary (pituitary, less often hypothalamic)
Causes	Autoimmune adrenalitis (commonest in the West; isolated or in autoimmune polyendocrine syndrome type 1 or the commoner type 2); tuberculosis (developing countries, as in Addison's era); cytomegalovirus in HIV, fungi; bilateral haemorrhage/infarction; infiltration (amyloid, sarcoid, haemochromatosis; metastasis rarely); surgical or pharmacological adrenalectomy (ketoconazole, CYP17 inhibitors such as abiraterone)	Tumour, radiation, autoimmunity, pituitary apoplexy (Sheehan syndrome peripartum), trauma, congenital ACTH deficiency; exogenous glucocorticoid suppression of the HPA axis (most relevant to surgeons); critical illness-related corticosteroid insufficiency
Mineralocorticoid	Deficient	Preserved — glomerulosa is ACTH independent
Skin	Hyperpigmentation (POMC peptides on MC1 receptor)	Alabaster pallor
Electrolytes	Hyponatraemia, hyperkalaemia , raised creatinine	Hyponatraemia (SIADH-type), no hyperkalaemia
Replacement	Glucocorticoid plus fludrocortisone	Glucocorticoid only

- **Postoperative day 1 cortisol <3.4 µg/dL** is common after unilateral adrenalectomy but of unclear significance; **replacement is largely unnecessary without preoperative hypercortisolism** — yet keep a high index of suspicion.
- Fig. 146.33 groups causes: primary (autoimmune, infective, infiltration, haemorrhage/infarction, bilateral or pharmacological adrenalectomy) versus secondary (pituitary, hypothalamic, congenital ACTH deficiency).

Clinical Characteristics

- **Chronic**: non-specific fatigue and anorexia for months.
- **Adrenal crisis**: **hypotension unresponsive to fluids**, often mistaken for an **acute abdomen** (pain, nausea, vomiting, fever with hypovolaemia); **hypoglycaemic seizures in children**.

- **After unilateral adrenalectomy** insufficiency is unlikely but possible — especially for **cortisol-secreting lesions** (suppressed contralateral gland) and after **previous contralateral partial or radical nephrectomy**, which may have damaged or removed the other adrenal → review old operative and pathology reports and imaging.

TABLE 146.14 CLINICAL MANIFESTATIONS OF ADRENAL INSUFFICIENCY (GROUPED BY DEFICIENT HORMONE)

Deficiency	Symptoms	Signs
Glucocorticoid	Fatigue, anorexia and weight loss (failure to thrive), gastric pain, nausea, vomiting, myalgia, arthralgia, dizziness	Fever; hypotension; hyponatraemia (SIADH); anaemia, lymphocytosis, eosinophilia ; hypoglycaemia; raised TSH [*] ; hypercalcaemia [*]
Mineralocorticoid (primary only)	Dizziness; salt craving [*]	Postural hypotension (systolic <100 mmHg); hyponatraemia; hyperkalaemia [*] ; raised creatinine [*]
Adrenal androgen (women, children)	Fatigue; dry itchy skin; loss of libido	Loss of axillary/pubertal hair; absent adrenarche/pubarche
POMC peptides	—	Hyperpigmentation [*] (excess) vs alabaster pallor (secondary; deficiency)

From Arlt W, Allolio B, 2003. ^{*} Primary adrenal insufficiency only. TSH, thyroid-stimulating hormone; SIADH, syndrome of inappropriate antidiuretic hormone secretion.

- **What it tests:** the features that exist only in primary disease — salt craving, hyperkalaemia, hyperpigmentation — are exactly those needing mineralocorticoid loss or high ACTH. **Trap:** hyponatraemia occurs in both, because glucocorticoid deficiency causes SIADH.

Diagnostic Tests

- **Clinical suspicion → morning cortisol and ACTH;** primary disease shows abnormal aldosterone and renin; **confirm with the corticotropin (ACTH stimulation) test;** most autoimmune cases have **anti-21-hydroxylase antibodies**.
- Secondary insufficiency needs specialist HPA-axis testing.

Treatment

- **Hydrocortisone or cortisone acetate**, with **half to two-thirds of the daily dose in the morning** and the rest in one or two later doses (circadian mimicry). **Fludrocortisone only in primary disease.** Androgen supplementation for persistent constitutional symptoms (some experts).
- **Perioperative stress dosing** is debated — usual dose is often enough, but coverage is given because the downside is negligible and crisis catastrophic. **Mineralocorticoid is unnecessary** for patients on chronic steroids (aldosterone intact).

TABLE 146.15 PERIOPERATIVE GLUCOCORTICOID ADMINISTRATION IN PATIENTS ON CHRONIC STEROID THERAPY

Surgical stress	Example	Glucocorticoid dose
Minor	Local anaesthetic, <1 h (inguinal hernia)	Hydrocortisone 25 mg or equivalent
Moderate	Lower-limb vascular surgery, joint replacement	Hydrocortisone 50–75 mg; e.g., usual prednisone (10 mg) + hydrocortisone 50 mg IV intraoperatively
Major	Oesophagogastrectomy, cardiopulmonary bypass	Usual dose (e.g., prednisone 40 mg or IV equivalent within 2 h pre-op) + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Data from Salem M et al., 1994; Axelrod L, 2003.

TABLE 146.16 RELATIVE POTENCY OF COMMON GLUCOCORTICOID PREPARATIONS

Agent	Biological half-life (h)	Anti-inflammatory potency	Physiological replacement (mg/day)
Hydrocortisone	8–12	1	20
Cortisone acetate	—	0.8	25
Prednisone	18–36	4	5
Prednisolone	—	4	5
Methylprednisolone	—	5	4
Triamcinolone	—	5	4
Dexamethasone	36–54	25–50	0.5

Modified from Krasner AS, 1999. — = not stated in the source.

- **Pattern:** potency and half-life rise together; **hydrocortisone 20 ≈ prednisone 5 ≈ dexamethasone 0.5 mg/day**. **Trap:** dexamethasone's long half-life and lack of detection in cortisol assays are what make it the dexamethasone suppression agent and the postoperative "test-friendly" replacement.

Prognosis

- Effect of primary insufficiency on life expectancy unknown, but **quality of life is impaired** (low energy, libido, psychological effects); **secondary (hypopituitary) insufficiency causes premature death**.

Summary

- In urology: surgical or pharmacological loss of adrenal tissue; high index of suspicion; **previous ipsilateral nephrectomy before contralateral surgery** is the classic at-risk setting; consider in patients on chronic steroids and the critically ill; **low threshold for endocrinology input**.

KEY POINTS: ADRENAL INSUFFICIENCY

- **Mineralocorticoid deficiency only in primary disease;** pituitary or hypothalamic failure (including chronic steroid suppression) spares aldosterone.
- **Prior partial or radical nephrectomy + contralateral renal or adrenal surgery = risk of postoperative adrenal crisis** — review old reports and imaging for the remaining adrenal.

Concept — *Whether insufficiency is primary or secondary decides whether aldosterone is lost, and that one fact separates the two clinically (hyperkalaemia, salt craving, pigmentation) and therapeutically (fludrocortisone). For the urologist, the danger is iatrogenic: the last functioning adrenal tissue is what an operation removes.*

Congenital Adrenal Hyperplasia

- **Autosomal recessive** steroidogenic enzyme defect with low cortisol; **21-hydroxylase deficiency in >95%**. Loss of negative feedback → **high ACTH → cortical hyperplasia and androgen excess**. Usually diagnosed in childhood (Chapter 59).
- Patients and **carriers** are prone to **benign cortical adenomas**, not carcinoma; **undiagnosed non-classic 21-hydroxylase deficiency** can present with adrenal neoplasms later in life.
- **No CAH screening for incidentaloma patients** — genetic screening has not shown excess undiagnosed CAH.

CONCLUSIONS

- The adrenals are **essential to life-sustaining homeostasis** yet, despite frequent incidental lesions, are **infrequent primary initiators of disease**.

- Significant abnormalities need a **systematic multidisciplinary approach**, in which the urologist — through minimally invasive and retroperitoneal expertise and the renal–adrenal link — naturally contributes evaluation and **primary surgical management**.